2. Digital Logic and Microprocessor

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Syllabus

2.1 Digital logic: Number Systems, Logic Levels, Logic Gates, Boolean algebra, Sum-of-Products Method, Product-of-Sums Method, Truth Table to Karnaugh Map. (AExE0201)

2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

2.3 Sequential logic circuit: RS Flip-Flops, Gated Flip-Flops, Edge Triggered Flip-Flops, Mater- Slave Flip-Flops. Types of Registers, Applications of Shift Registers, Asynchronous Counters, Synchronous Counters. (AExE0203)

2.4 Microprocessor: Internal Architecture and Features of microprocessor, Assembly Language Programming. (AExE0204)

2.5 Microprocessor system: Memory Device Classification and Hierarchy, Interfacing I/O and Memory Parallel Interface. Introduction to Programmable Peripheral Interface (PPI), Serial Interface, Synchronous and Asynchronous Transmission, Serial Interface Standards. Introduction to Direct Memory Access (DMA) and DMA Controllers. (AExE0205)

2.6 Interrupt operations: Interrupt, Interrupt Service Routine, and Interrupt Processing. (AExE0206)

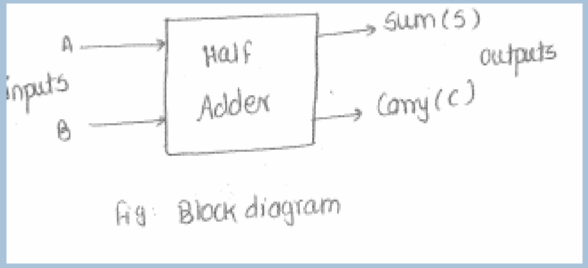
2.2 Combinational and arithmetic circuits: Multiplexetures, Demultiplexetures, Decoder, Encoder, Binary Addition, Binary Subtraction, operation on Unsigned and Signed Binary Numbers. (AExE0202)

ADDER - HALF

- Binary adders nay be of two types:
  - Half Adder
  - Full Adder

Half Adder:

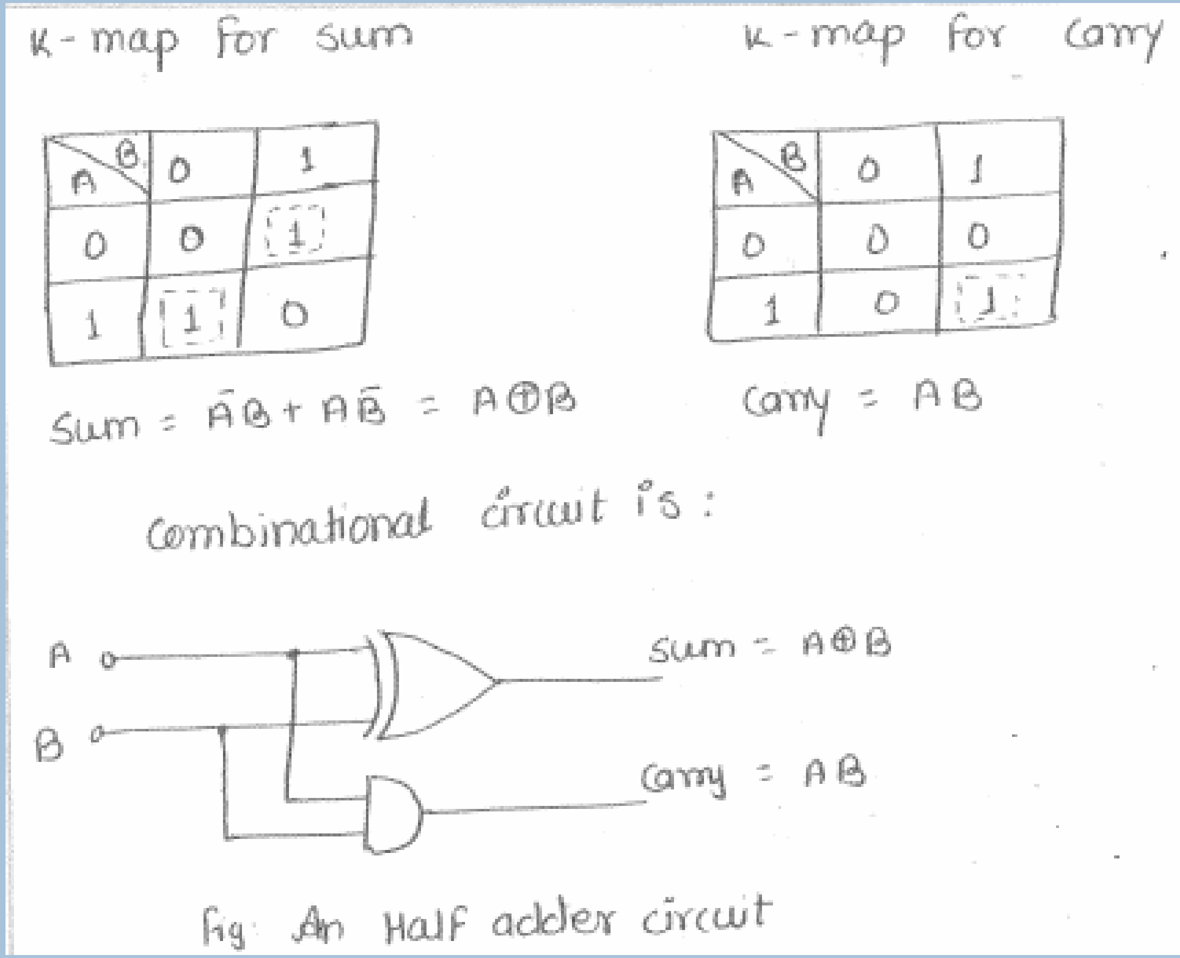
- Half adder is a combinational logic circuit with two inputs and two outputs.
- It is the basic building block for addition of two single bit numbers
- This circuit has two outputs namely, Sum and Carry.

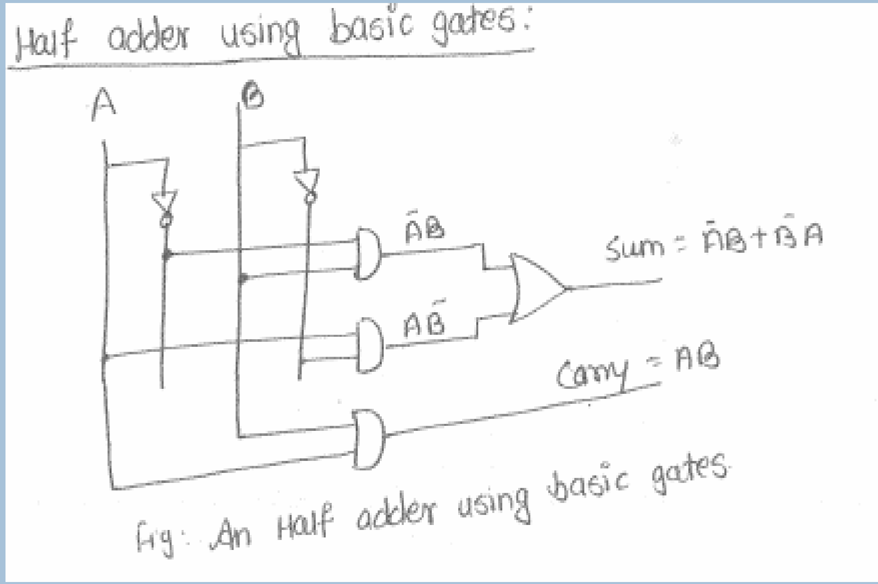


Truth table:

Inputs		Outputs	
A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

ADDER - HALF



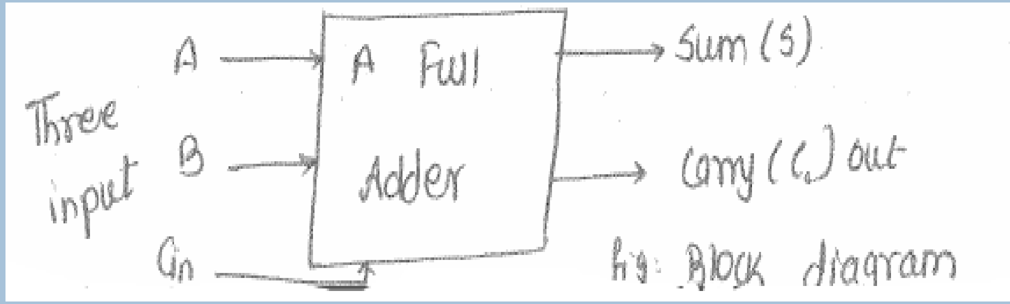


- Limitations:
  - The addition of three bits is not possible to perform by using an half adder circuit.

ADDER - FULL

Full Adder:

- To overcome the drawback of an half adder circuit, a 3-single bit adder circuit called full adder is developed.
- Basically, a full adder is a three input and two output combinational circuit.



- Application: A full adder acts as the basic building block of the 4 bit/ 8 bit binary/ BCD adder Ics such as 7483.

Truth table:

Inputs			**Outputs**	
A	B	C_{in}	Sum	Carry (C_o)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

ADDER - FULL

K-map for sum (S)

A \ BC_{in}	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$S = A'B'C_{in} + A'BC_{in}' + AB'C_{in}' + ABC_{in}$$

$$= A \oplus B \oplus C_{in}$$

K-map for carry out (C_o)

A \ BC_{in}	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$$C_o = AC_{in} + BC_{in} + AB$$

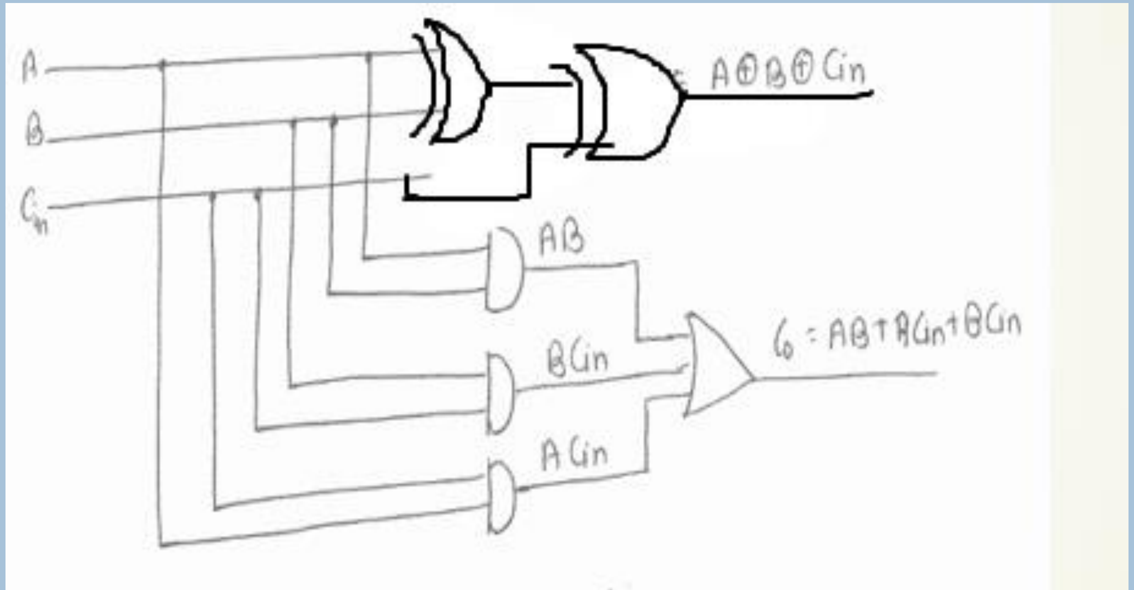


fig: A full adder circuit

SUBTRACTOR - HALF

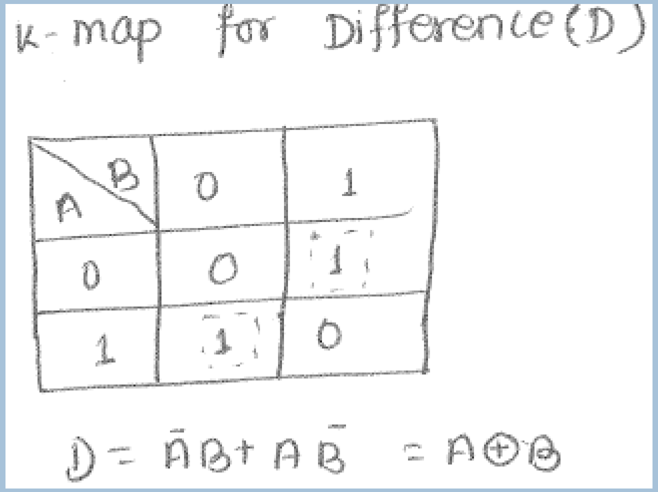
- Binary Subtractor nay be of two types:
  - Half Subtractor
  - Full Subtractor

Half Subtractor:

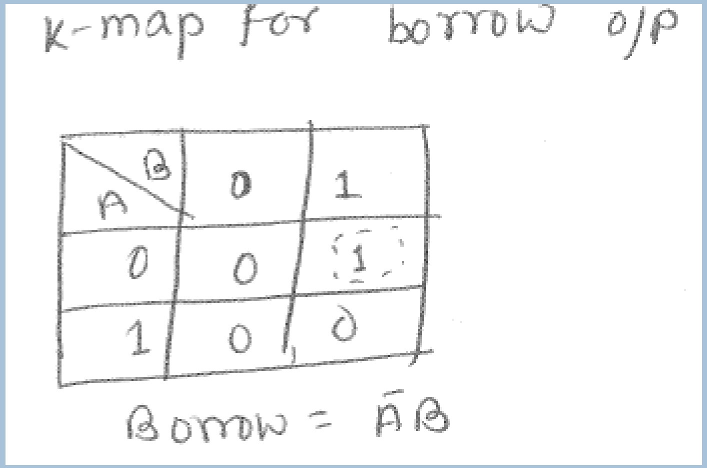
- Half subtractor may be defined as combinational circuit with two inputs and two outputs (i.e. difference and borrow)
- In subtraction (A-B), A is called minuend bit and B is called as Subtrahend bit.

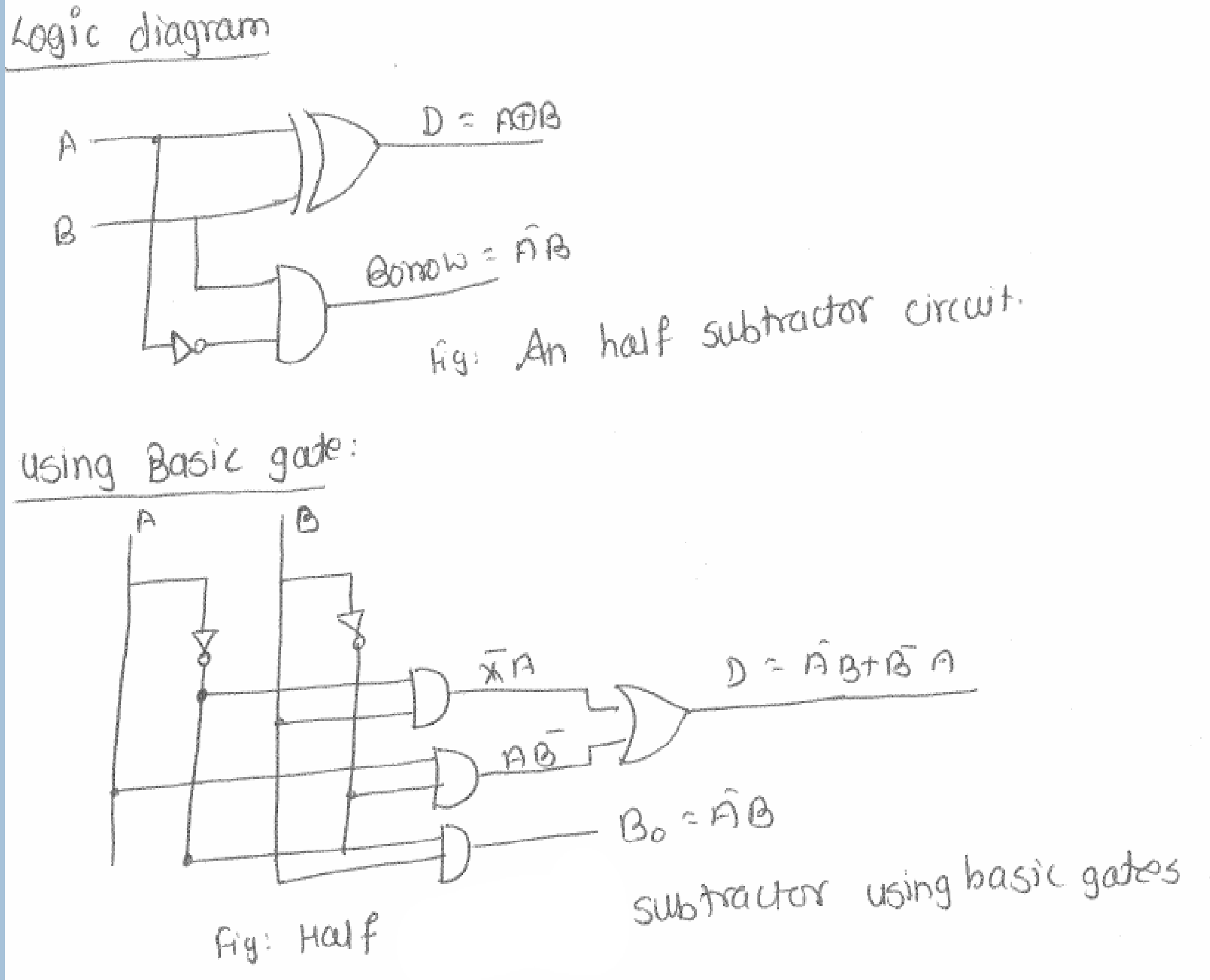
- Truth table:

Inputs		Outputs	
A	B	Difference (A-B)	Borrow (B_o)
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0



SUBTRACTOR - HALF





SUBTRACTOR - FULL

Full Subtractor:

- A Full Subtractor is a combinational circuit with three inputs A, B, B_{in} and two outputs D (Difference) and Borrow ($B_{o.}$)
- Here A is the minuend, B is the subtrahend, B_{in} is the borrow produced by the previous stage,

D is the difference output and B_o is the borrow output.

Truth table:

Inputs			**Outputs**	
A	B	B_{in}	Difference (A-B-B_{in})	Borrow (B_o)
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

SUBTRACTOR - FULL

K-map

For difference output

A \ B Bin	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$D = \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + ABB_{in}$$

$$= B_{in}(\bar{A}\bar{B} + AB) + \bar{B}_{in}(\bar{A}B + A\bar{B})$$

$$= B_{in}(\overline{A \oplus B}) + \bar{B}_{in}(A \oplus B)$$

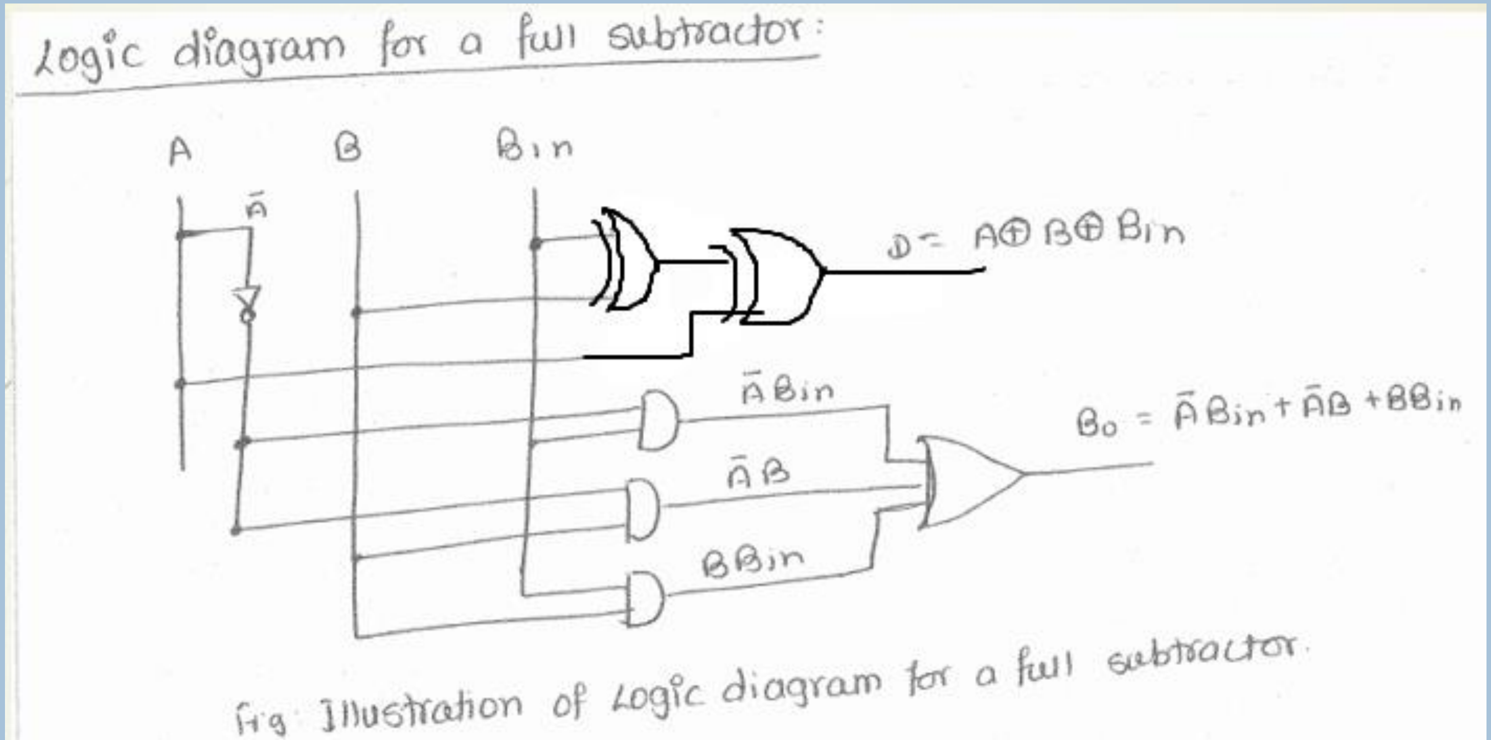
$$= A \oplus B \oplus B_{in}$$

for Borrow output

A \ B Bin	00	01	11	10
0	0	1	1	1
1	0	0	1	0

$$B_0 = \bar{A}B_{in} + \bar{A}B + BB_{in}$$

SUBTRACTOR - FULL



BINARY PARALLEL ADDER

- A full adder is capable of adding only two single digit binary numbers along with a carry input.
- But, in practice, we need to add binary numbers which are much longer than just one bit.
- To add two n- bit binary numbers, we need to use the n-bit parallel adder.
- It makes use of a number of full adders in cascade.
- The carry output of the previous full adder is connected to the carry input of the next full adder.

A 4-bit Binary Parallel Adder:

Input Carry	0	1	1	0	C_i
Augend	1	0	1	1	A_i
Addend	0	0	1	1	B_i
Sum	1	1	1	0	S_i
output carry	0	0	1	1	C_{i+1}

BINARY PARALLEL ADDER

- A binary parallel adder is a digital function that produces the arithmetic sum of two binary numbers in parallel.
- It consists of full-adder connected in cascade, with the output carry from one full adder connected to the input carry of the next full-adder.

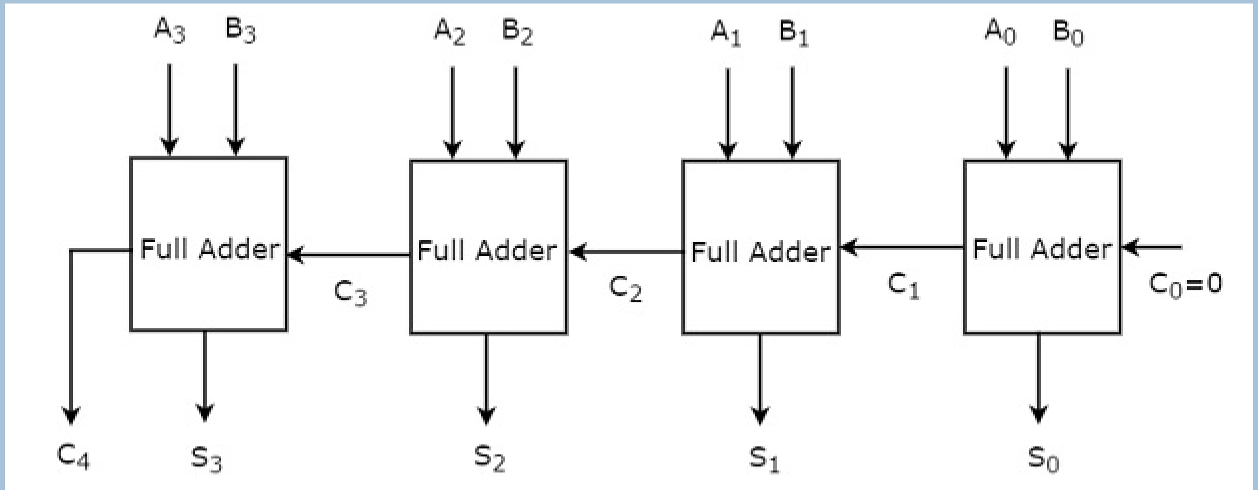
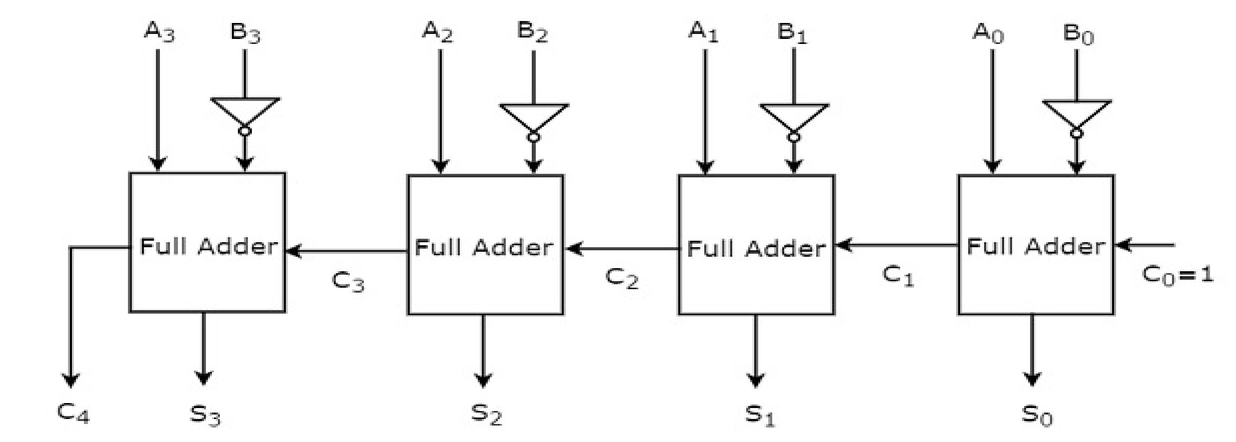


Fig: 4-bit Binary Parallel adder

4-BIT BINARY PARALLEL SUBTRACTOR

- The 4-bit binary subtractor produces the subtraction of two 4-bit numbers.
- Let the 4 bit binary numbers, $A=A_3A_2A_1A_0$ and $B=B_3B_2B_1B_0$.
- Internally, the operation of 4-bit Binary subtractor is similar to that of 4-bit Binary adder.
- If the normal bits of binary number A, complemented bits of binary number B and initial carry borrow, C_{in} as one are applied to 4-bit Binary adder, then it becomes 4-bit Binary subtractor.
- The block diagram of 4-bit binary subtractor is shown in the following figure.



Note : A-B = A+ B' +1

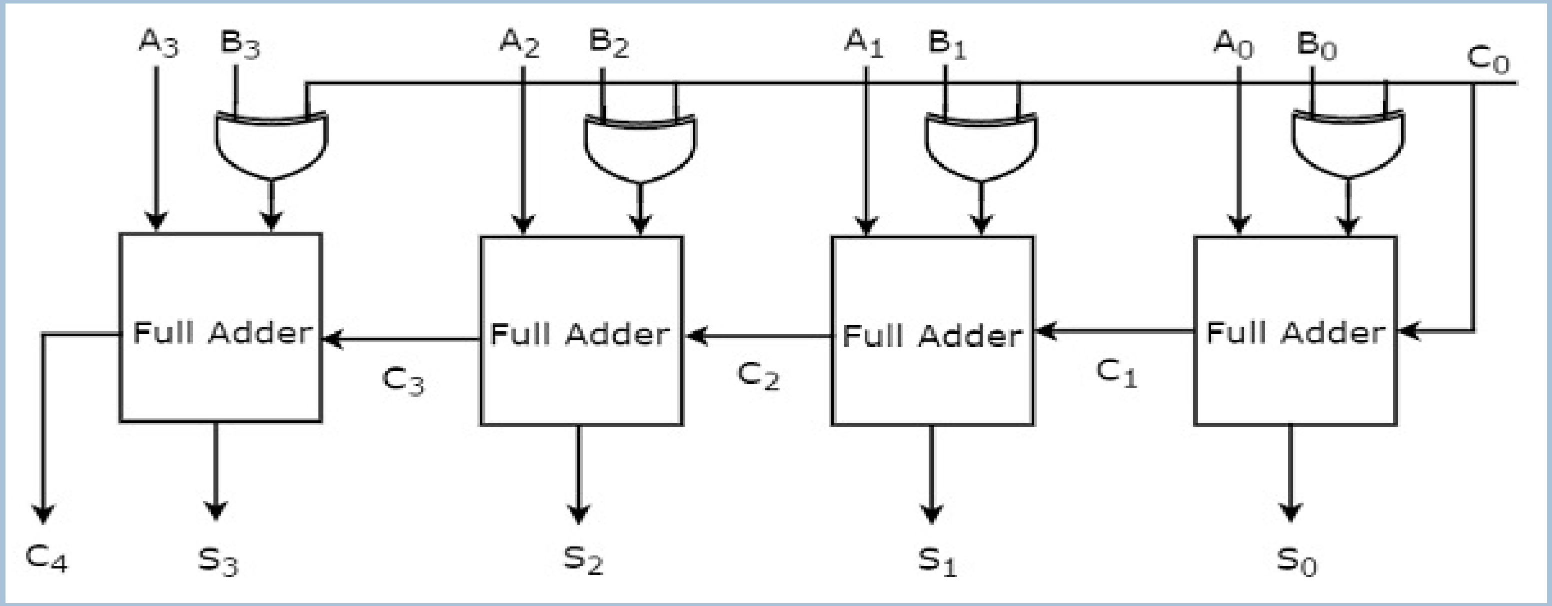
4-BIT BINARY PARALLEL SUBTRACTOR

- This 4-bit binary subtractor produces an output, which is having at most 5 bits.
- If Binary number A is greater than Binary number B, then MSB of the output is zero and the remaining bits hold the magnitude of A-B.
- If Binary number A is less than Binary number B, then MSB of the output is one. So, take the 2's complement of output in order to get the magnitude of A-B

4-BIT BINARY PARALLEL ADDER/SUBTRACTOR

- The circuit, which can be used to perform either addition or subtraction of two binary numbers at any time is known as Binary Adder / subtractor.
- Both, Binary adder and Binary subtractor contain a set of Full adders, which are cascaded.
- The input bits of binary number A are directly applied in both Binary adder and Binary subtractor.
- The input bits of binary number B are directly applied to Full adders in Binary adder, whereas the complemented bits of binary number B are applied to Full adders in Binary subtractor.
- The initial carry, $C_0 = 0$ is applied in 4-bit Binary adder, whereas the initial carry borrow, $C_0 = 1$ is applied in 4-bit Binary subtractor.
- We know that a 2-input Ex-OR gate produces an output, which is same as that of first input when other input is zero. Similarly, it produces an output, which is complement of first input when other input is one.

4-BIT BINARY PARALLEL ADDER/SUBTRACTOR



= 0 for adder

=1 for subtractor

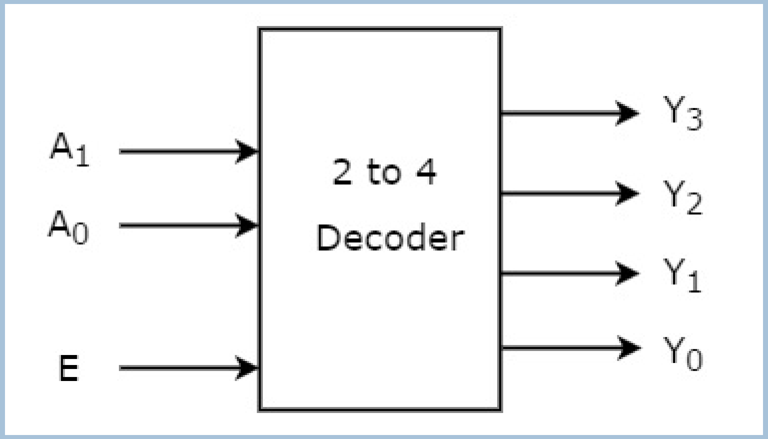
- If initial carry, C_0 is zero, then each full adder gets the normal bits of binary numbers A & B. So, the 4-bit binary adder / subtractor produces an output, which is the addition of two binary numbers A & B.
- If initial borrow, C_0 is one, then each full adder gets the normal bits of binary number A & complemented bits of binary number B. So, the 4-bit binary adder / subtractor produces an output, which is the subtraction of two binary numbers A & B.

DECODERS

- Decoder is a combinational circuit that has ‘n’ input lines and maximum of 2^n output lines.
- One of these outputs will be active High based on the combination of inputs present, when the decoder is enabled.
- That means decoder detects a particular code. The outputs of the decoder are nothing but the min terms of ‘n’ input variables lines, when it is enabled.

2 to 4 Decoder

- Let 2 to 4 Decoder has two inputs A1 & A0 and four outputs Y3, Y2, Y1 & Y0. The block diagram of 2 to 4 decoder is shown in the following figure.



DECODERS

One of these four outputs will be '1' for each combination of inputs when enable, E is '1'. The Truth table of 2 to 4 decoder is shown below.

Enable	Inputs		Outputs			
E	A_1	A_0	Y_3	Y_2	Y_1	Y_0
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

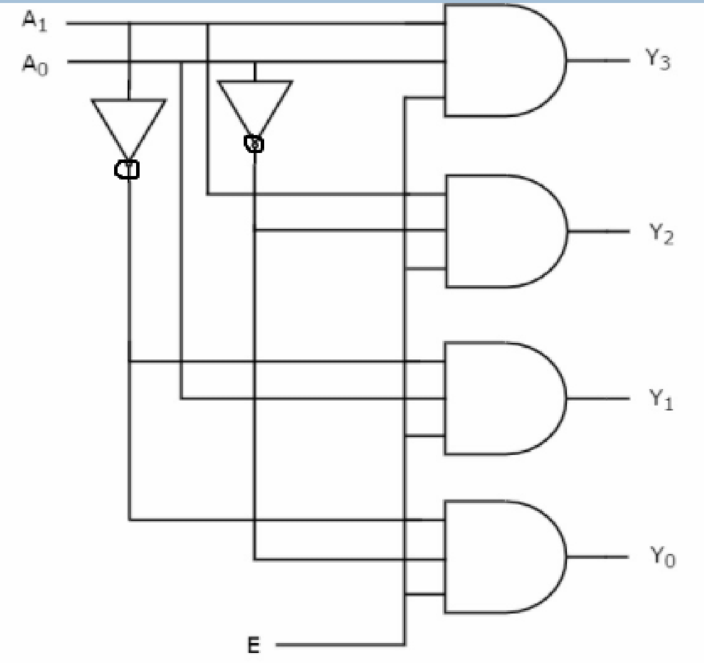
From Truth table, we can write the **Boolean functions** for each output as

$$Y_3 = E.\,A_1.\,A_0$$

$$Y_2 = E.\,A_1.\,A_0'$$

$$Y_1 = E.\,A_1'.\,A_0$$

$$Y_0 = E.\,A_1'.\,A_0'$$



DECODERS

3 X 8 DECODER:

Truth Table

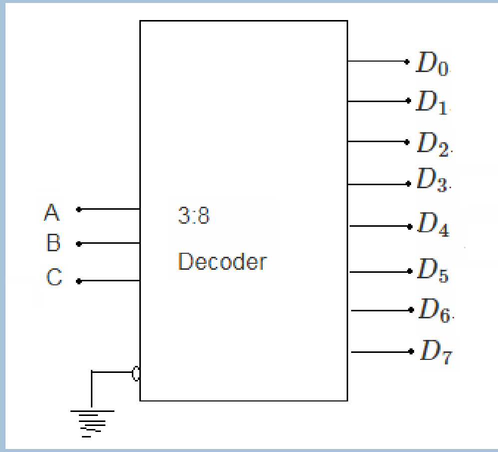
A	B	C	D0	D1	D2	D3	D4	D5	D6	D7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Function

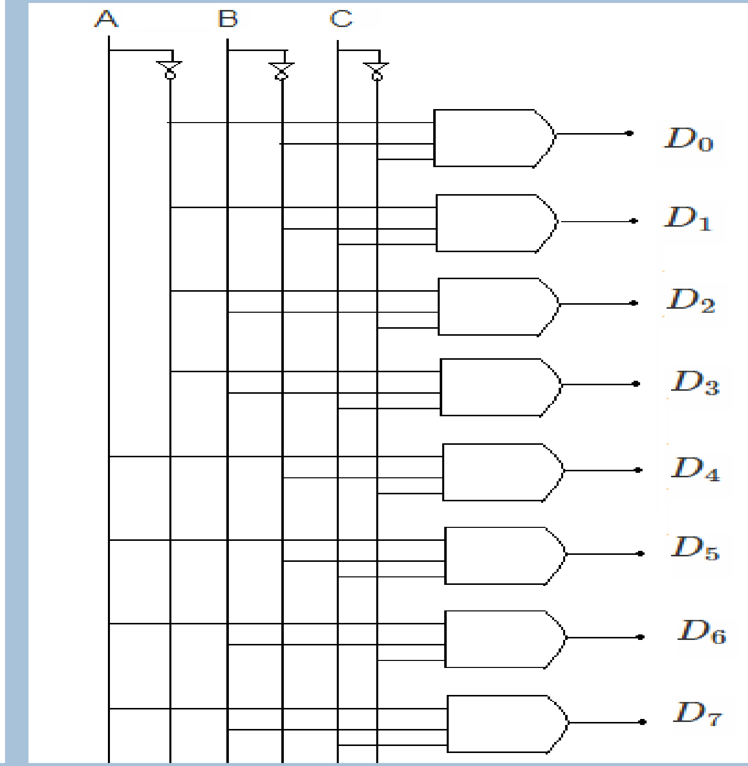
$D_0 = \bar{A}\bar{B}\bar{C},\quad D_1 = \bar{A}\bar{B}C,\quad D_2 = \bar{A}B\bar{C},$

$D_3 = \bar{A}BC,\quad D_4 = A\bar{B}\bar{C},\quad D_5 = A\bar{B}C,$

$D_6 = AB\bar{C},\quad D_7 = ABC$



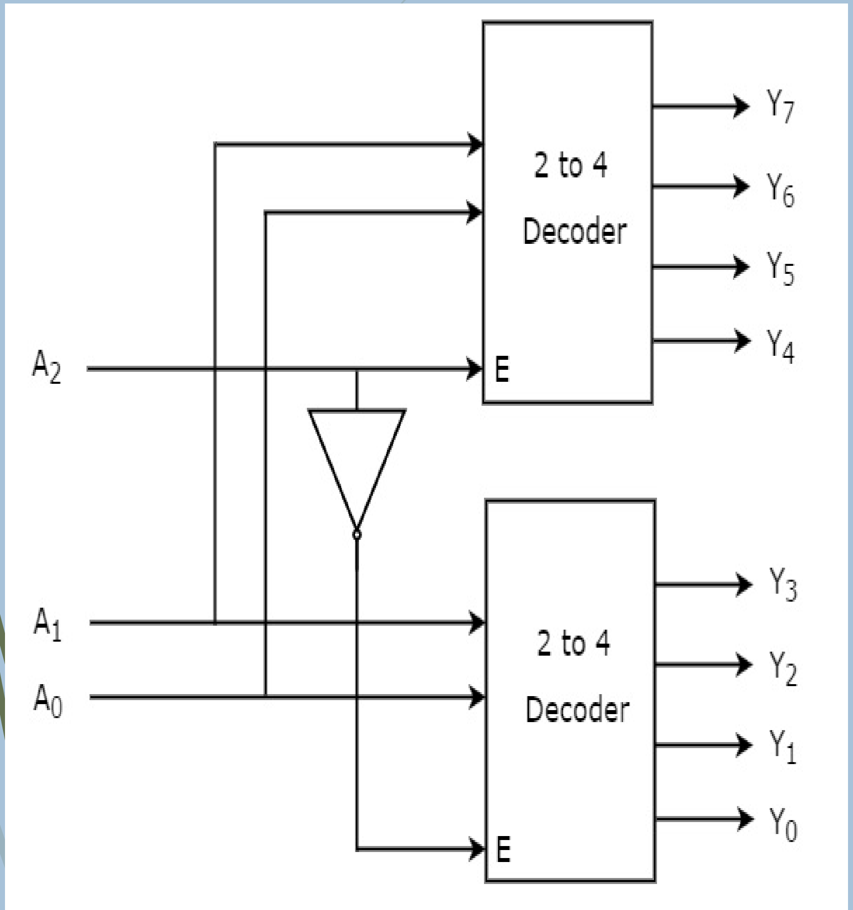
Block Diagram



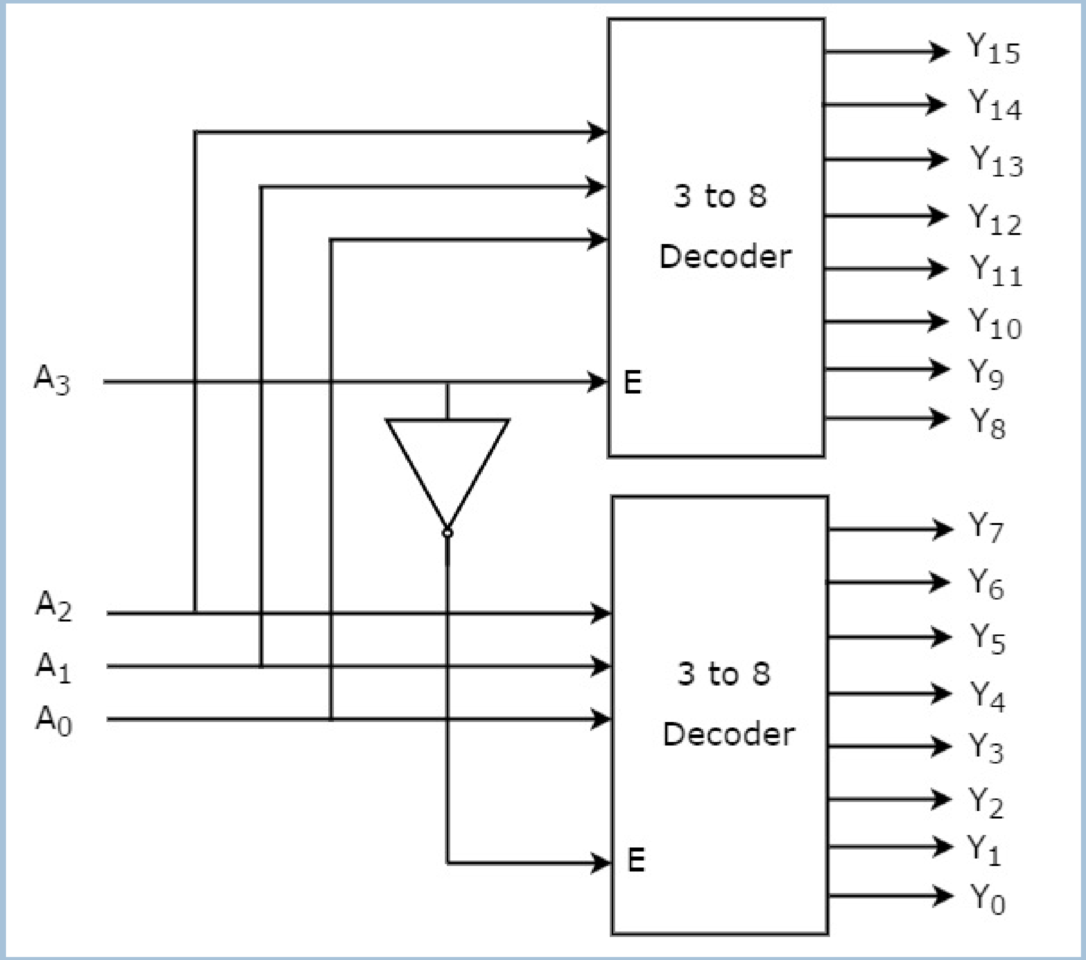
Logic Diagram

IMPLEMENTATION OF HIGHER ORDER DECODERS

3 to 8 decoder using 2 to 4 decoder

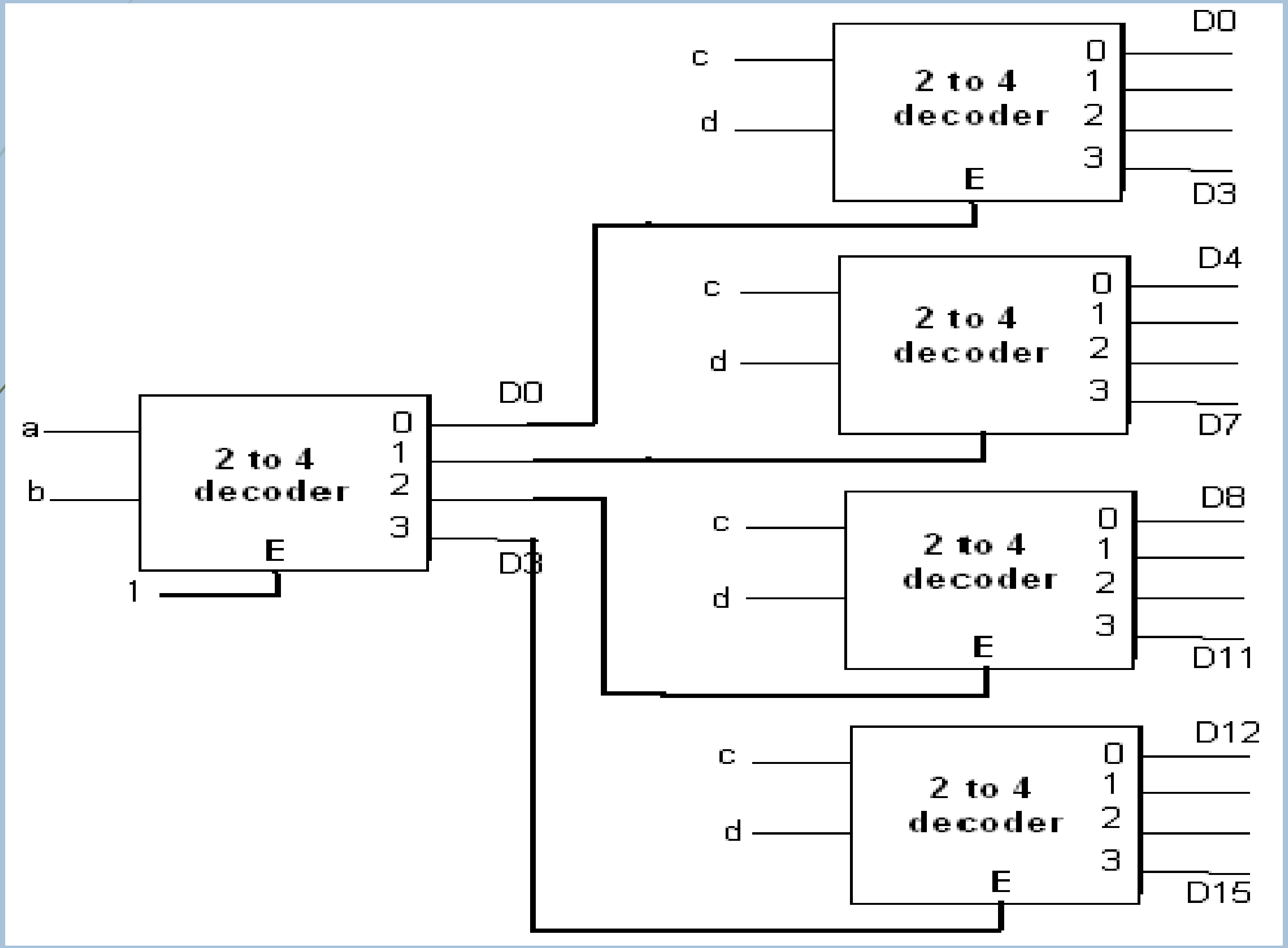


4 to 16 decoder using 3 to 8 decoder



IMPLEMENTATION OF HIGHER ORDER DECODERS

4 to 16 decoder using 2 to 4 decoder

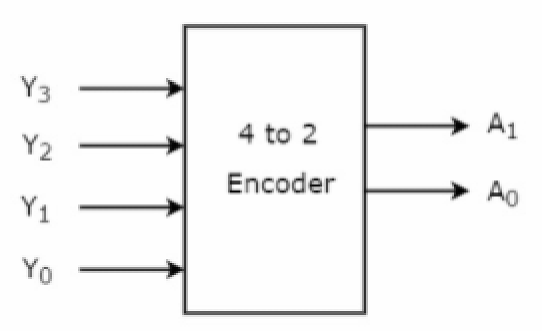


ENCODERS

- An Encoder is a combinational circuit that performs the reverse operation of Decoder.
- It has maximum of 2^n input lines and 'n' output lines.
- It will produce a binary code equivalent to the input, which is active High.
- Therefore, the encoder encodes 2^n input lines with 'n' bits. It is optional to represent the enable signal in encoders.

4 to 2 Encoder

- Let 4 to 2 Encoder has four inputs Y_3, Y_2, Y_1 & Y_0 and two outputs A_1 & A_0. The block diagram of 4 to 2 Encoder is shown in the following figure.



ENCODERS

At any time, only one of these 4 inputs can be ‘1’ in order to get the respective binary code at the output.

The Truth table of 4 to 2 encoder is shown below.

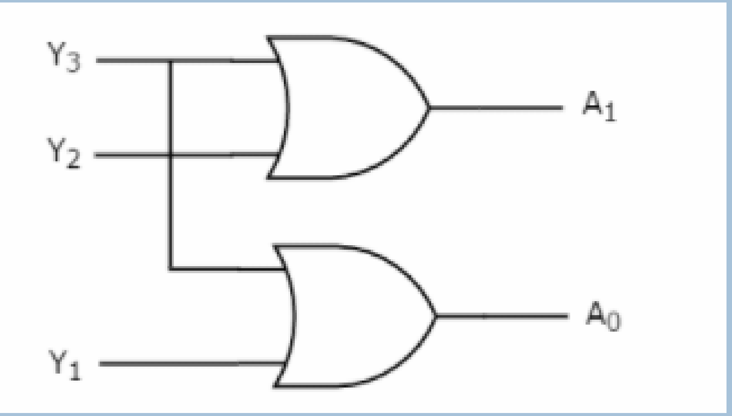
Inputs				Outputs	
Y_3	Y_2	Y_1	Y_0	A_1	A_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

From Truth table, we can write the **Boolean functions** for each output as

$$A_1 = Y_3 + Y_2$$

$$A_0 = Y_3 + Y_1$$

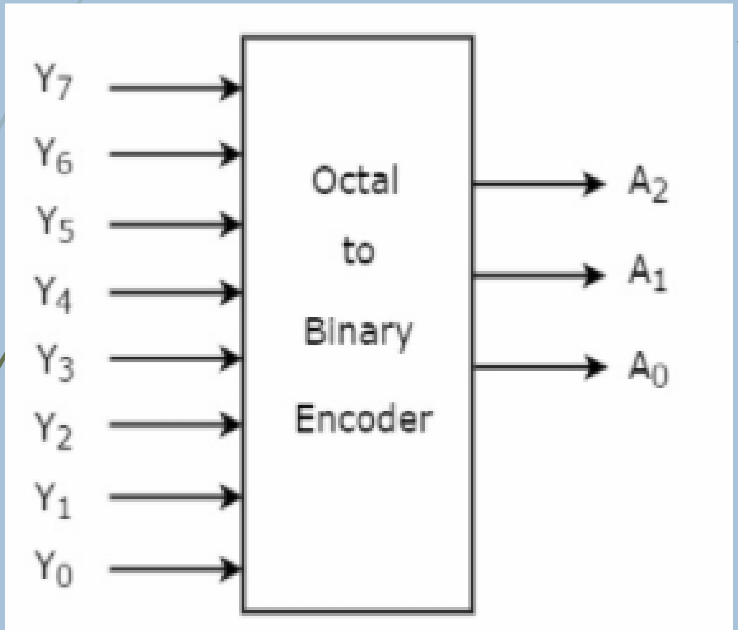
Logic Diagram



ENCODERS – OCTAL TO BINARY ENCODER

Octal to binary Encoder has eight inputs, Y7 to Y0 and three outputs A2, A1 & A0.

- Octal to binary encoder is nothing but 8 to 3 encoder. The block diagram of octal to binary Encoder is shown in the following figure



Truth Table and Function

Inputs								Outputs		
Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

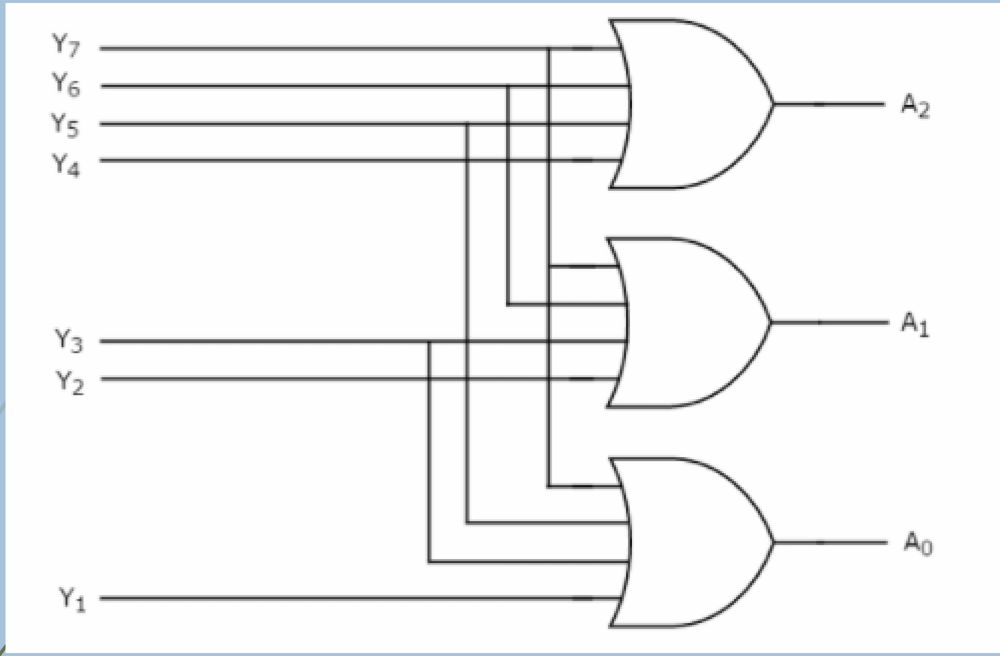
From Truth table, we can write the **Boolean functions** for each output as

$$A_2 = Y_7 + Y_6 + Y_5 + Y_4$$

$$A_1 = Y_7 + Y_6 + Y_3 + Y_2$$

$$A_0 = Y_7 + Y_5 + Y_3 + Y_1$$

ENCODERS – OCTAL TO BINARY ENCODER



Logic Diagram

Drawbacks of Encoder

- There is an ambiguity, when all outputs of encoder are equal to zero
- If more than one input is active High, then the encoder produces an output, which may not be the correct code.

ENCODERS – PRIORITY ENCODER

We considered one more output, V in order to know, whether the code available at outputs is valid or not.

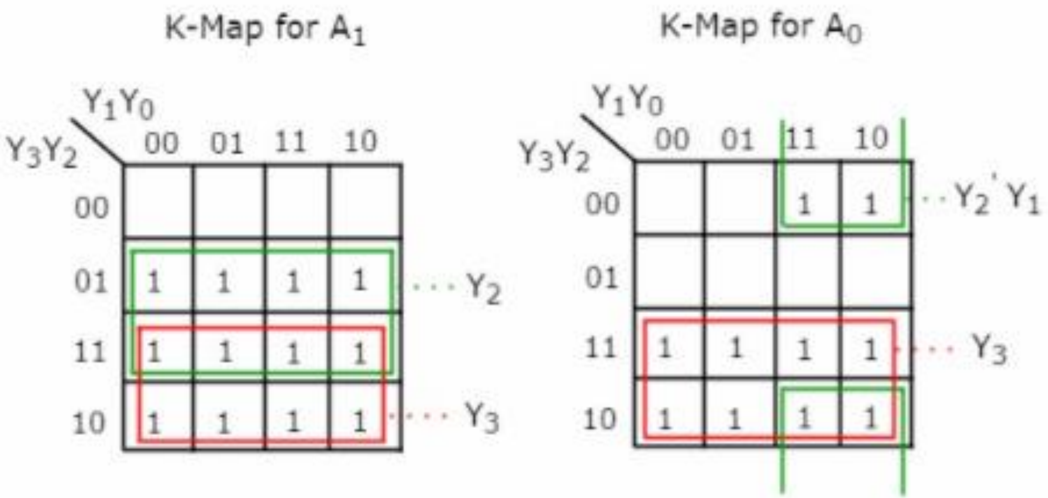
- If at least one input of the encoder is ‘1’, then the code available at outputs is a valid one. In this case, the output, V will be equal to 1.
- If all the inputs of encoder are ‘0’, then the code available at outputs is not a valid one. In this case, the output, V will be equal to 0.

Truth Table

Inputs				Outputs		
Y_3	Y_2	Y_1	Y_0	A_1	A_0	V
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1

ENCODERS – PRIORITY ENCODER

K-MAP and Function



The simplified **Boolean functions** are

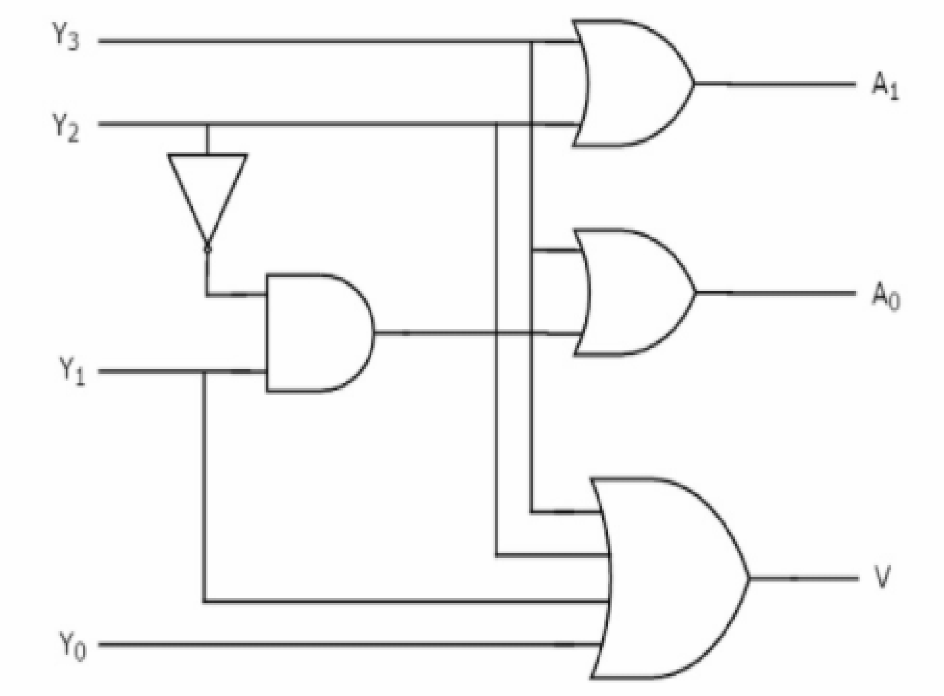
$$A_1 = Y_3 + Y_2$$

$$A_0 = Y_3 + Y_2'Y_1$$

Similarly, we will get the Boolean function of output, V as

$$V = Y_3 + Y_2 + Y_1 + Y_0$$

Logic Diagram

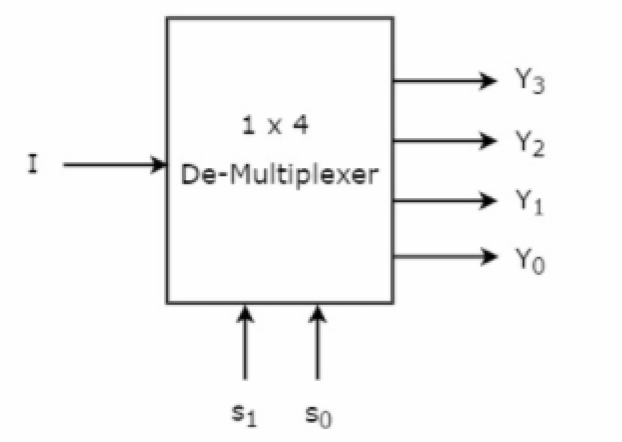


DEMULTIPLEXERS (DEMUX)

- A demultiplexer (or demux) is a device that takes a single input line and routes it to one of several digital output lines.
- A demultiplexer of 2^n outputs has n select lines, which are used to select which output line to send the input.
- A demultiplexer is also called a data distributor.

1x4 De-Multiplexer

- 1x4 De-Multiplexer has one input I, two selection lines, s1 & s0 and four outputs Y3, Y2, Y1 &Y0. The block diagram of 1x4 De-Multiplexer is shown in the following figure.



DEMULTIPLEXERS (DEMUX)

Truth Table

Selection Inputs		Outputs			
S_1	S_0	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	I
0	1	0	0	I	0
1	0	0	I	0	0
1	1	I	0	0	0

From the above Truth table, we can directly write the **Boolean functions** for each output as

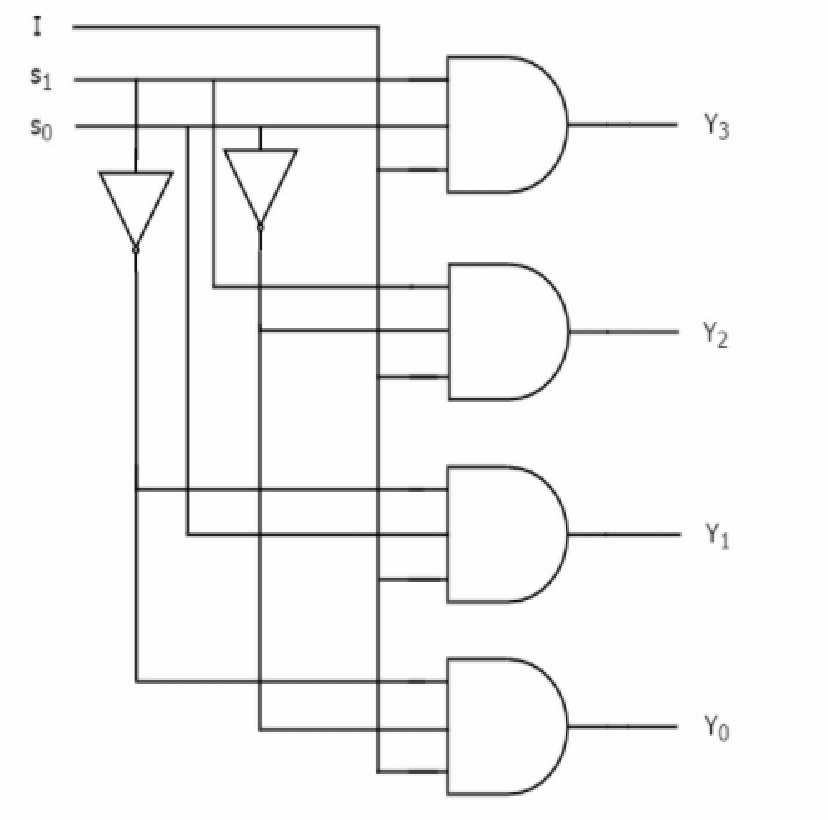
$$Y_3 = s_1 s_0 I$$

$$Y_2 = s_1 s_0' I$$

$$Y_1 = s_1' s_0 I$$

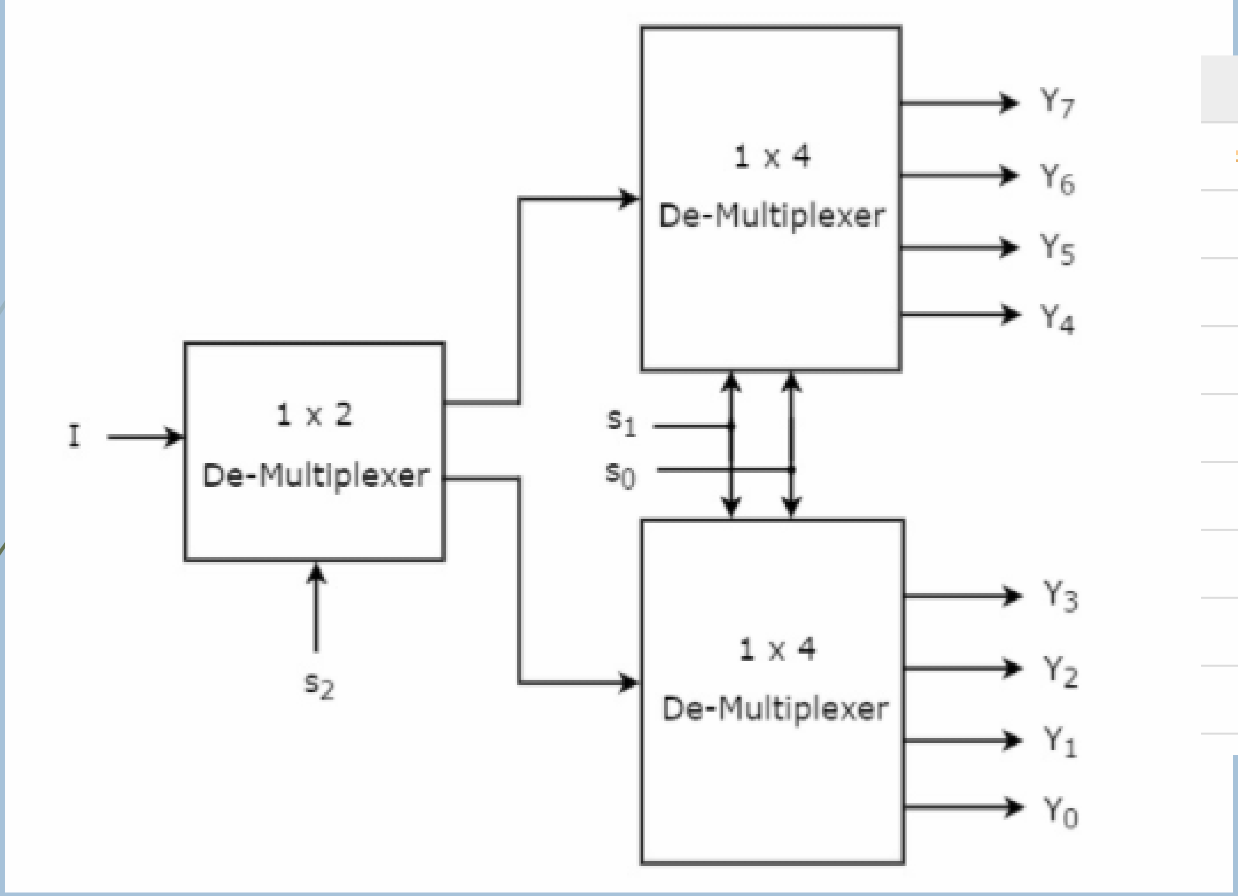
$$Y_0 = s_1' s_0' I$$

Logic Diagram



IMPLEMENTATION OF HIGHER ORDER DE-MULTIPLEXERS

1 X 8 DE-MUX using 1 X 4 and 1 X 2 DE-MUX



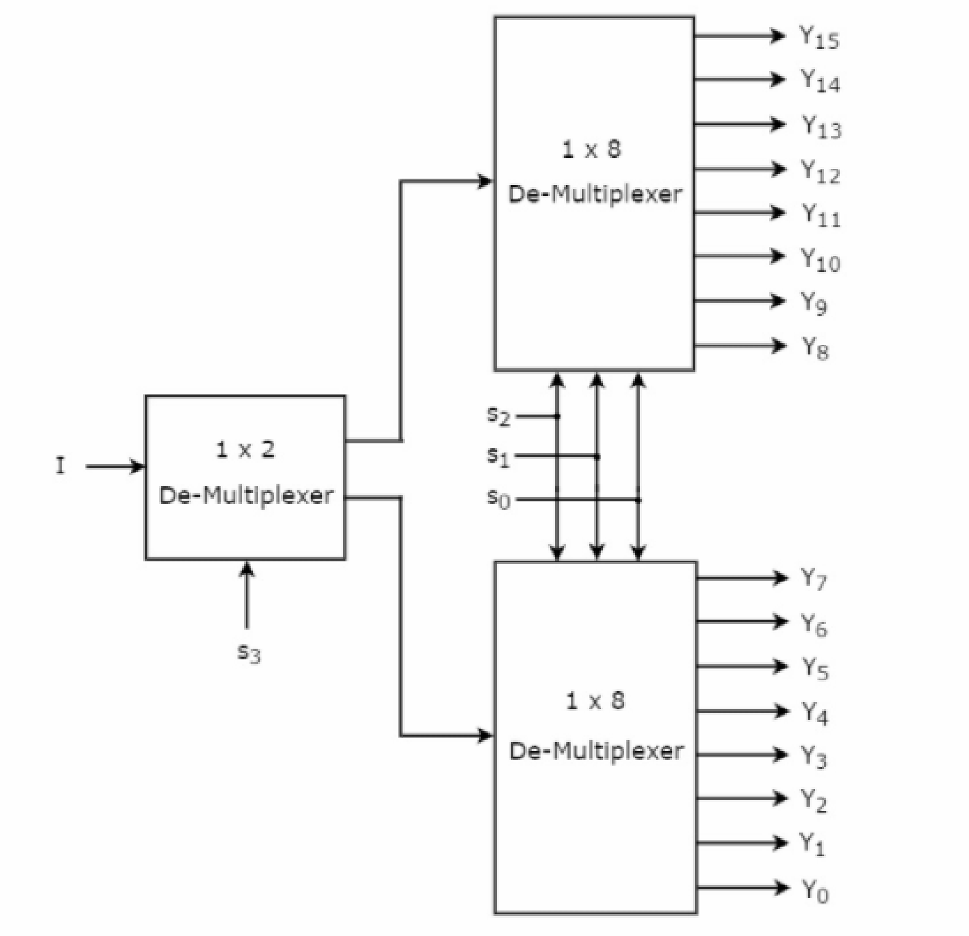
Logic Diagram

Truth Table

Selection Inputs			Outputs							
s_2	s_1	s_0	Y_7	Y_6	Y_5	Y_4	Y_3	Y_2	Y_1	Y_0
0	0	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	1	0	0	0	0	0	0	1	0	0
0	1	1	0	0	0	0	1	0	0	0
1	0	0	0	0	0	1	0	0	0	0
1	0	1	0	0	1	0	0	0	0	0
1	1	0	0	1	0	0	0	0	0	0
1	1	1	1	0	0	0	0	0	0	0

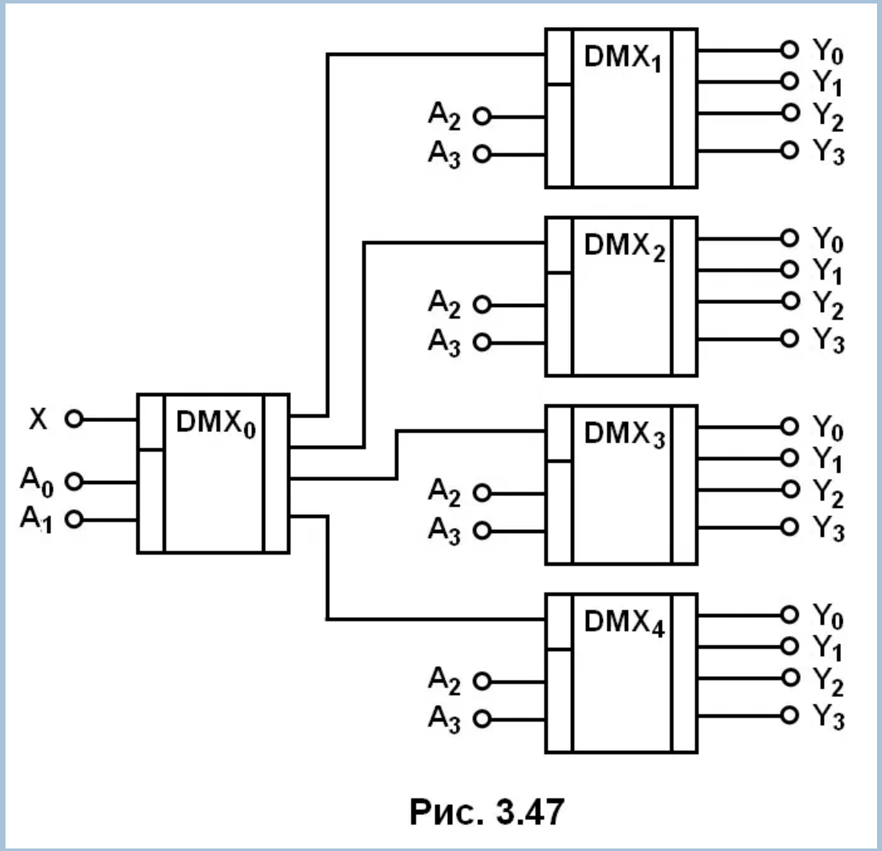
IMPLEMENTATION OF HIGHER ORDER DE-MULTIPLEXERS

1 X 16 DE-MUX using 1 x 8 and 1 x 2 DE-MUX



Logic Diagram

1 X 16 DE-MUX using 1 x 4



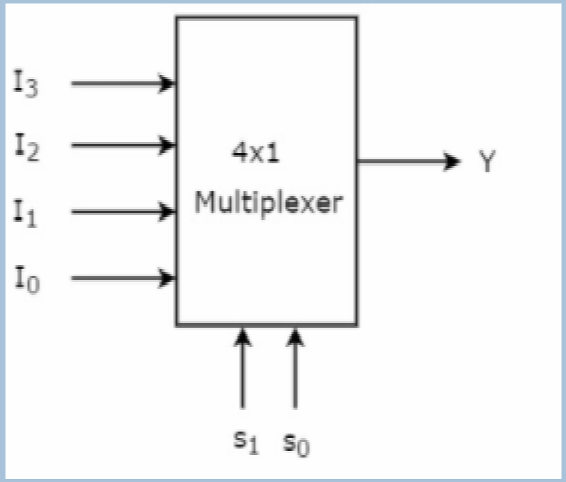
X = INPUT, A_3,A_2,A_1,A_0 are selection line

MULTIPLEXERS (MUX)

- Multiplexer is a combinational circuit that has maximum of 2^n data inputs, 'n' selection lines and single output line.
- One of these data inputs will be connected to the output based on the values of selection lines.
- Since there are 'n' selection lines, there will be 2^n possible combinations of zeros and ones.
- So, each combination will select only one data input. Multiplexer is also called as Mux.

4x1 Multiplexer

- 4x1 Multiplexer has four data inputs I_3, I_2, I_1 & I_0, two selection lines s_1 & s_0 and one output Y. The block diagram of 4x1 Multiplexer is shown in the following figure.



MULTIPLEXERS (MUX)

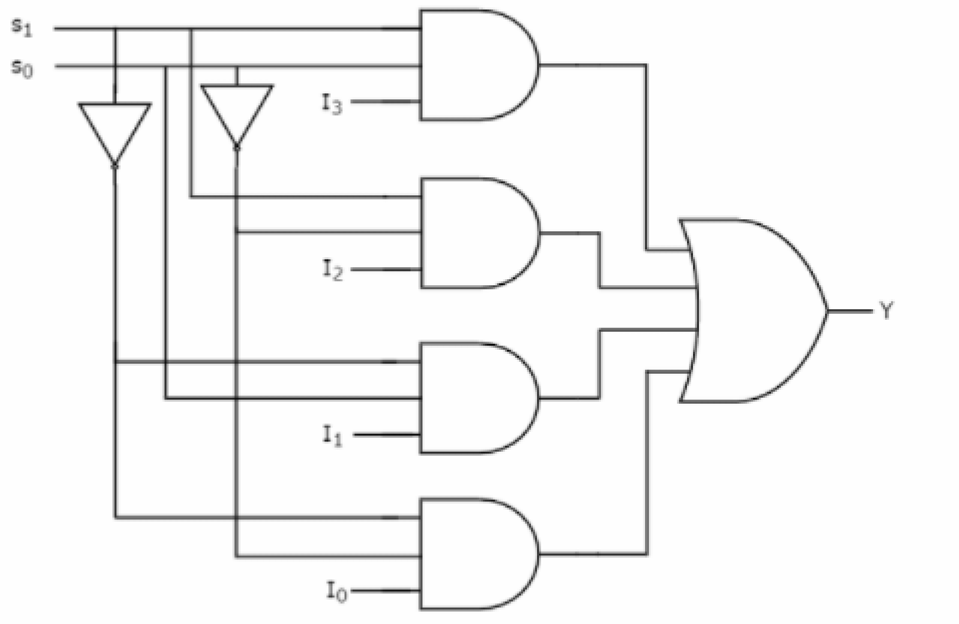
Truth table of 4x1 Multiplexer is shown below.

Selection Lines		Output
S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

From Truth table, we can directly write the **Boolean function** for output, Y as

$$Y = S_1{}'S_0{}'I_0 + S_1{}'S_0I_1 + S_1S_0{}'I_2 + S_1S_0I_3$$

Logic Diagram



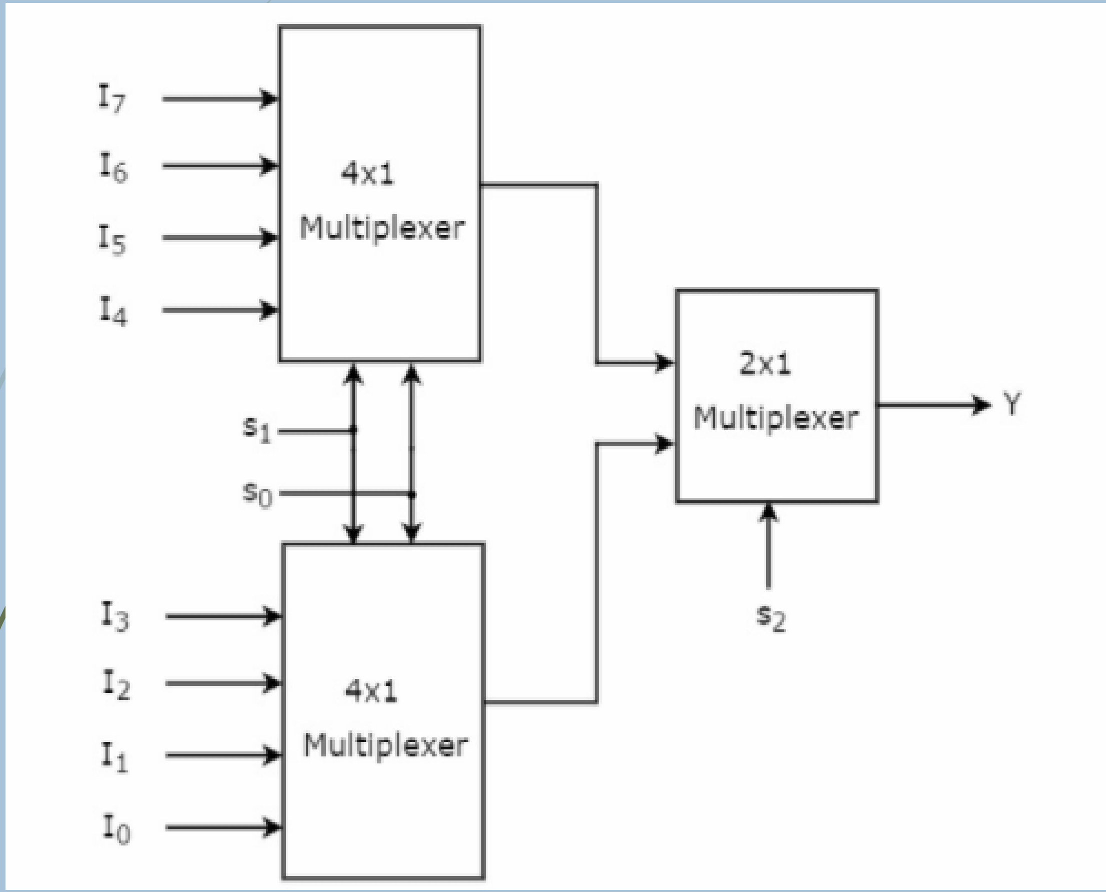
MULTIPLEXERS (MUX)

Application

- Communication system
- Telephone network
- Computer memory
- Transmission from the computer system of a satellite
- Data acquisition system

IMPLEMENTATION OF HIGHER ORDER MULTIPLEXERS

8 X 1 MUX using 4 x 1 and 2 x 1 MUX

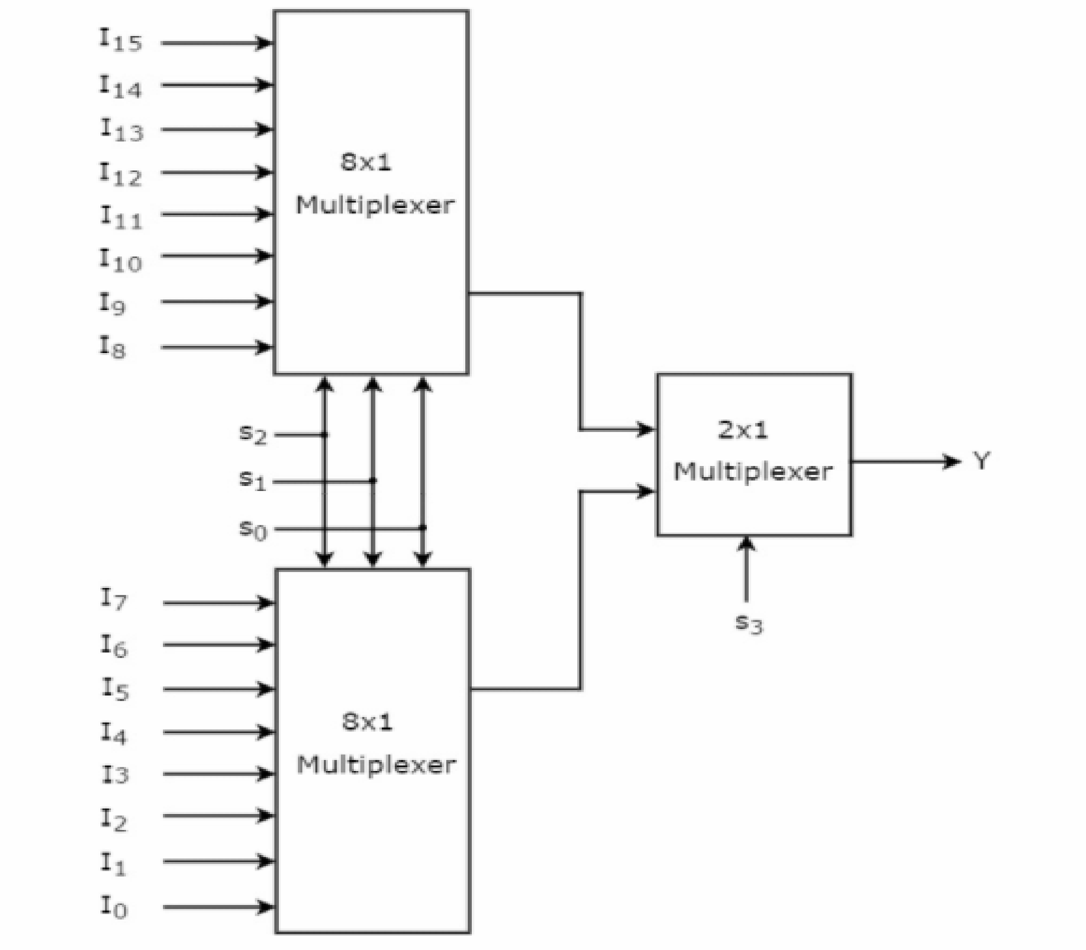


Truth Table

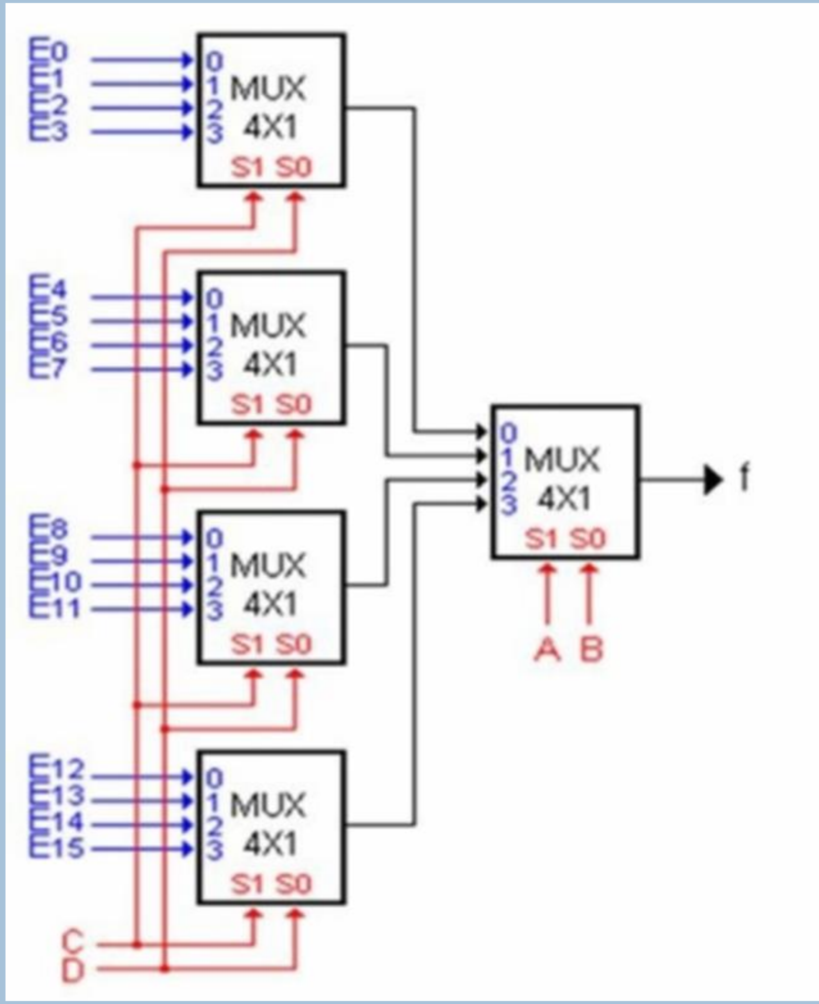
Selection Inputs			Output
S_2	S_1	S_0	Y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

IMPLEMENTATION OF HIGHER ORDER MULTIPLEXERS

16 X 1 MUX using 8 x 1 and 2 x 1 MUX



16 X 1 MUX using 4 x 1



BINARY ADDITION

Binary Addition Rules

		Carry Over	Result
1.	0 + 0	0	0
2.	0 + 1	0	1
3.	1 + 0	0	1
4.	1 + 1	1	0
5.	1 + 1 + 1	1	1

BINARY SUBTRACTION

Rules of Binary Subtraction

$\Rightarrow 1 - 0 = 1$

$\Rightarrow 1 - 1 = 0$

$\Rightarrow 0 - 0 = 0$

$\Rightarrow 0 - 1 = 1$

(This can not be done directly, hence we borrow one digit from the digit to the left or the next higher order digit.)

SIGNED AND UNSIGNED BINARY NUMBERS

Unsigned Numbers

As we already know, the unsigned numbers don't have any sign for representing negat numbers. So the unsigned numbers are always positive. By default, the decimal numb representation is positive. We always assume a positive sign in front of each decimal digit.

Signed Numbers

The signed numbers have a sign bit so that it can differentiate positive and negative integer numbers. The signed binary number technique has both the sign bit and the magnitude of the number. For representing the negative decimal number, the corresponding symbol in front of the binary number will be added.

SIGNED AND UNSIGNED BINARY NUMBERS

1. **Sign-Magnitude form**

   In this form, a binary number has a bit for a sign symbol. If this bit is set to 1, the number will be negative else the number will be positive if it is set to 0. Apart from this sign-bit, the n-1 bits represent the magnitude of the number.

2. **1's Complement**

   By inverting each bit of a number, we can obtain the 1's complement of a number. The negative numbers can be represented in the form of 1's complement. In this form, the binary number also has an extra bit for sign representation as a sign-magnitude form.

3. **2's Complement**

   By inverting each bit of a number and adding plus 1 to its least significant bit, we can obtain the 2's complement of a number. The negative numbers can also be represented in the form of 2's complement. In this form, the binary number also has an extra bit for sign representation as a sign-magnitude form.

Combinational and Arithmetic Circuits

1 Which of the following is correct for multiplexer?

a) Several inputs and single output

b) Single input and several outputs

c) Single input and single output

d) Several inputs and several outputs

2 TDM stands for ____________________

a) Time direct measurement

b) Time division multiplexing

c) Time direct multiplexing

d) Time division measurement

Combinational and Arithmetic Circuits

3 . Multiplexers work with ____________

a) Analog signal

b) Digital signal

c) Both analog and digital signal

d) None of the mentioned

4 . Which of the following represent multiple input single output switch?

a) Multiplexer

b) De multiplexer

c) Both multiplexer and demultiplexer

d) None of the mentioned

Combinational and Arithmetic Circuits

5 Which of the following is analogous to multiplexer?

a) Data selector

b) Data multiplexer

c) Data filter

d) None of the mentioned

6 . Schematic symbol of multiplexer is _______________

a) Isosceles triangle

b) Isosceles trapezoid

c) Equilateral triangle

d) Rectangle

Combinational and Arithmetic Circuits

7 In digital multiplexer selector line is _______________

a) Analog value

b) Digital value

c) Unpredictable

d) None of the mentioned

8 . Which of the following is not a multiplexer?

a) 8-to-1 line

b) 16-to-1 line

c) 4-to-1 line

d) 1-to-4 line

Combinational and Arithmetic Circuits

9 What is a multiplexer?

a) It is a type of decoder which decodes several inputs and gives one output

b) A multiplexer is a device which converts many signals into one

c) It takes one input and results into many output

d) It is a type of encoder which decodes several inputs and gives one output

10 . The enable input is also known as __________

a) Select input

b) Decoded input

c) Strobe

d) Sink

Combinational and Arithmetic Circuits – SET B

11 . Which combinational circuit is renowned for selecting a single input from multiple inputs & directing the binary information to output line?

a) Data Selector

b) Data distributor

c) Both data selector and data distributor

d) DeMultiplexer

12 What is the function of an enable input on a multiplexer chip?

a) To apply Vcc

b) To connect ground

c) To active the entire chip

d) To active one half of the chip

Combinational and Arithmetic Circuits

13 Which is the major functioning responsibility of the multiplexing combinational circuit?

a) Decoding the binary information

b) Generation of all minterms in an output function with OR-gate

c) Generation of selected path between multiple sources and a single destination

d) Encoding of binary information

14 One multiplexer can take the place of __________

a) Several SSI logic gates

b) Combinational logic circuits

c) Several Ex-NOR gates

d) Several SSI logic gates or combinational logic circuits

Combinational and Arithmetic Circuits

15 . A digital multiplexer is a combinational circuit that selects ___________

a) One digital information from several sources and transmits the selected one

b) Many digital information and convert them into one

c) Many decimal inputs and transmits the selected information

d) Many decimal outputs and accepts the selected information

16 In a multiplexer, the selection of a particular input line is controlled by ___________

a) Data controller

b) Selected lines

c) Logic gates

d) Both data controller and selected lines

Combinational and Arithmetic Circuits

17 If the number of n selected input lines is equal to 2^m then it requires ____ select lines.

a) 2

b) m

c) n

d) 2^n

18 . How many select lines would be required for an 8-line-to-1-line multiplexer?

a) 2

b) 4

c) 8

d) 3

19 . How many NOT gates are required for the construction of a 4-to-1 multiplexer?

a) 3

b) 4

c) 2

d) 5

Combinational and Arithmetic Circuits

20 On subtracting (01010)2 from (11110)2 using 1's complement, we get ___________

a) 01001

b) 11010

c) 10101

d) 10100

21 . On subtracting (001100)2 from (101001)2 using 2's complement, we get ___________

a) 1101100

b) 011101

c) 11010101

d) 11010111

22 On addition of +38 and -20 using 2's complement, we get ___________

a) 11110001

b) 100001110

c) 010010

d) 110101011

Combinational and Arithmetic Circuits

23 On subtracting (010110)2 from (1011001)2 using 2's complement, we get ____________

a) 0111001

b) 1100101

c) 0110110

d) 1000011

24 On addition of 28 and 18 using 2's complement, we get ____________

a) 00101110

b) 0101110

c) 00101111

d) 1001111

25 On addition of -46 and +28 using 2's complement, we get ____________

a) -10010

b) -00101

c) 01011

d) 0100101

Combinational and Arithmetic Circuits

26 A decoder converts n inputs to __________ outputs

a) n

b) n^2

c) 2^n

d) n^n

27 Which of the following can be represented for decoder?

a) Sequential circuit

b) Combinational circuit

c) Logical circuit

d) None of the mentioned

28 Invalid BCD can be made to valid BCD by adding with ____________

a) 0101

b) 0110

c) 0111

d) 1001

Combinational and Arithmetic Circuits

29 Which of the following are building blocks of encoders

a) NOT gate

b) OR gate

c) AND gate

d) NAND gate

30 BCD to seven segment conversion is a _______________

a) Decoding process

b) Encoding process

c) Comparing process

d) None of the mentioned

31 Decoder is constructed from _________________

a) Inverters

b) AND gates

c) Inverters and AND gates

d) None of the mentioned

Combinational and Arithmetic Circuits

32. Which of the following represents a number of output lines for a decoder with 4 input lines?

a) 15

b) 16

c) 17

d) 18

33. A 4 to 2 encoder requires ___ number of logic gates?

- 2
- 3
- 4
- 5

34. An encoder generates ___ type code on each input?

- Hexa
- Binary
- Octal
- ASCII

Combinational and Arithmetic Circuits

35

Which of the following is the output "A,B" for an encoder with 4bits "Y1,Y2,Y3,Y4" as " 0010"?

- 00
- 01
- 10
- 11

36

). Which of the following is the output for 8 to 3 type encoder for input D [7] to [0] "00000001"?

- XXX
- 000
- 001
- 010

37

. Which of the following is the output of 3to8 type decoder when input is 1,100?

- A *B bar * C bar
- A * B* C
- A+B bar +C
- A bar +B +C

Combinational and Arithmetic Circuits

38 AND gate EXOR gate combination is ____

1. both full and half adder.
2. full adder
3. half adder
4. flip flop

39 In a half adder, the carry output is high if the inputs are:

1. 1, 1
2. 0, 0
3. 0, 1
4. 1, 0

40 If the inputs are P, Q and R, then in the full adder, find the output expression of the sum.

1. P OR Q OR R
2. P XOR Q XOR R
3. P OR Q AND R
4. P AND Q AND R

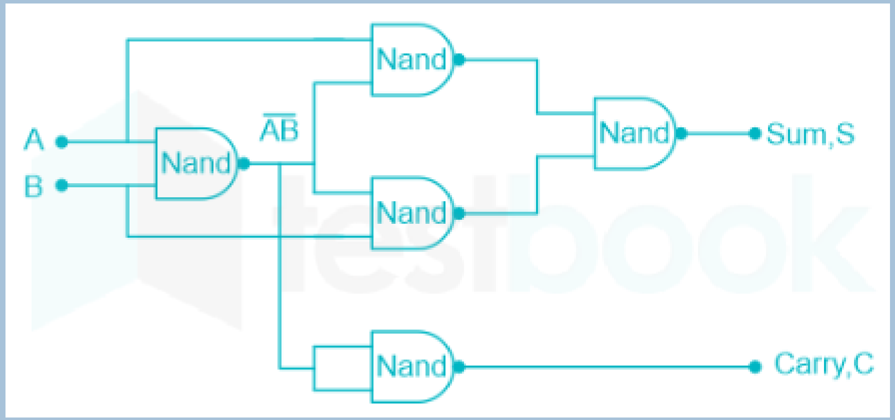
41 In full adder, there are

1. Two binary number inputs and two outputs
2. Three binary digit inputs and two binary digit outputs
3. Three binary digit inputs and three binary digit outputs
4. NAND and OR gates

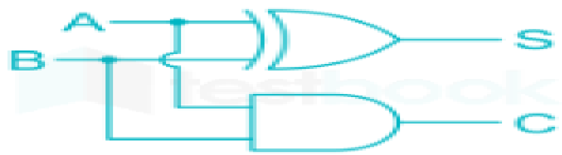
Combinational and Arithmetic Circuits

42 How many number of 2-input NAND gates are required to realise a half adder circuit?

1. 5
2. 6
3. 4
4. 8



43 The following circuit is a :



1. Half subtractor
2. Half adder
3. Full adder
4. Full subtractor

44 In half adder, the total number of inputs and outputs are:

1. 1, 2
2. 2, 1
3. 3, 2
4. 2, 2

Combinational and Arithmetic Circuits

45 For a full Adder

1. Sum = AB ⊕ BC ⊕ CA
   Carry = A, B, C
2. Sum = A, B, C
   Carry = A ⊕ B ⊕ C
3. Sum = A ⊕ B ⊕ C
   Carry = A, B, C
4. Sum = A ⊕ B ⊕ C
   Carry = AB + BC + AC

46 In which operation carry is obtained?

a) Subtraction
b) Addition
c) Multiplication
d) Both addition and subtraction

47 Half-adders have a major limitation in that they cannot ________

a) Accept a carry bit from a present stage
b) Accept a carry bit from a next stage
c) Accept a carry bit from a previous stage
d) Accept a carry bit from the following stages

48 If A, B and C are the inputs of a full adder then the carry is given by ________

a) A AND B OR (A OR B) AND C
b) A OR B OR (A AND B) C
c) (A AND B) OR (A AND B)C
d) A XOR B XOR (A XOR B) AND C

Combinational and Arithmetic Circuits

49

A half subtractor is an arithmetic circuit which performs subtraction operation on ______ input bits.

1. two
2. three
3. four
4. one

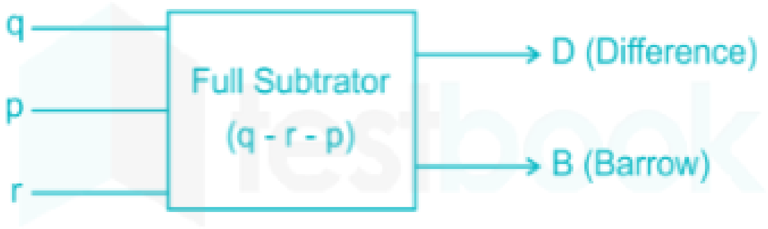
50

In a half-subtractor circuit with X and Y as inputs, the Borrow (M) and Difference (N = X - Y) are given by

1. $M = X \oplus Y, N = XY$
2. $M = XY, N = X \oplus Y$
3. $M = \bar{X}Y, N = X \oplus Y$
4. $M = X\bar{Y}, N = \overline{X \oplus Y}$

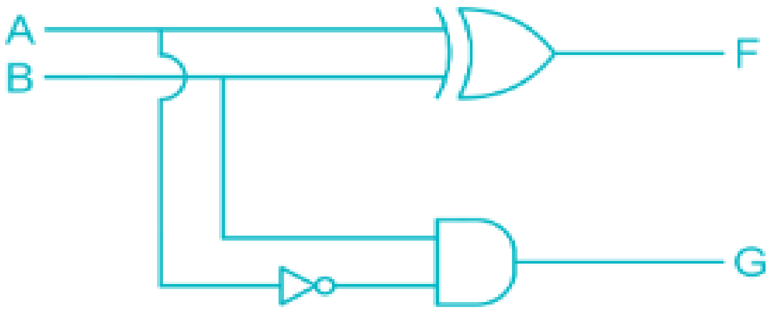
Combinational and Arithmetic Circuits

51 Find the Boolean expression for Borrow (B) as the circuit shown below



1. $\bar{p}q + \bar{p}r + qr$
2. $pq + pr + qr$
3. $p\bar{q} + pr + \bar{q}r$
4. $p\bar{q} + \bar{p}r + \bar{q}r$

52

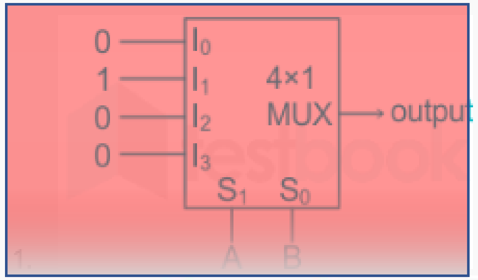


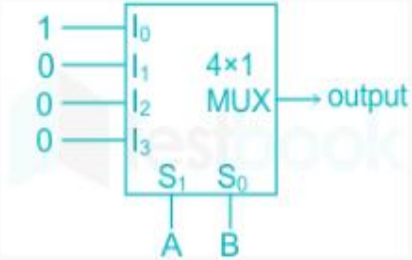
What is the circuit?

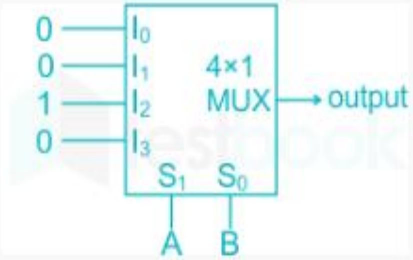
1. Half adder
2. Parity generator
3. Code convertor
4. Half subtractor

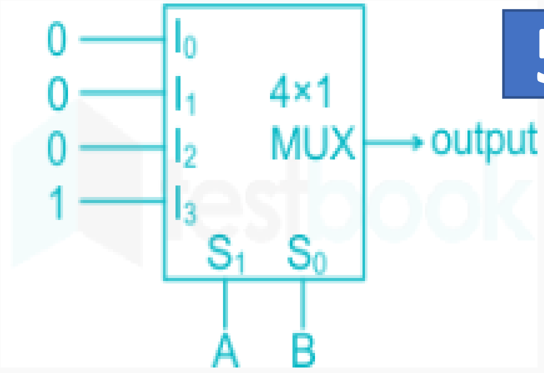
Combinational and Arithmetic Circuits

53 Which of the following circuit diagrams represents borrow of half subtractor?

1.


2.


3.


4.


54 For subtracting 1 from 0, we use to take a _____ from neighbouring bits.

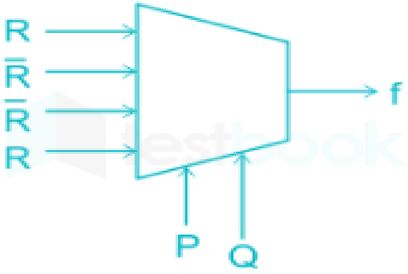
a) Carry

b) Borrow

c) Input

d) Output

55 The Boolean expression for the output f of the multiplexer shown b



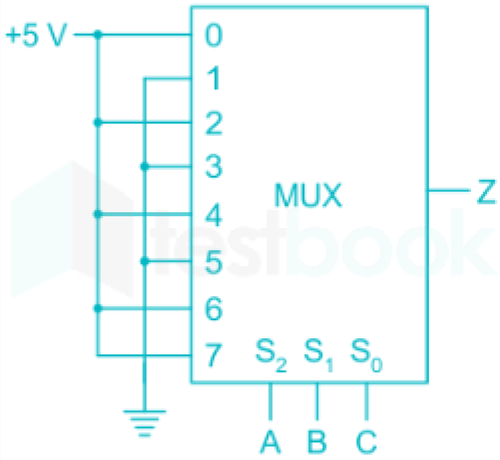
1. $\overline{P \oplus Q \oplus R}$
2. $P \oplus Q \oplus R$
3. $P + Q + R$
4. More than one of the above

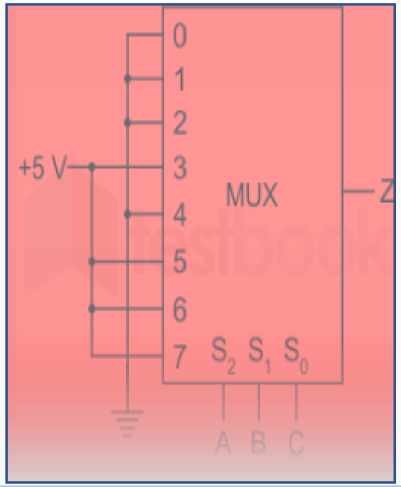
Combinational and Arithmetic Circuits

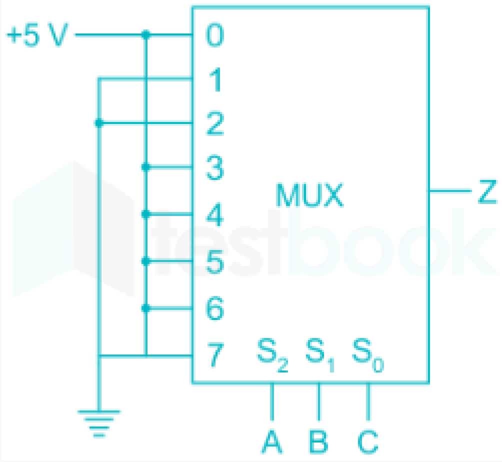
56 The device which changes from serial data to parallel data is:

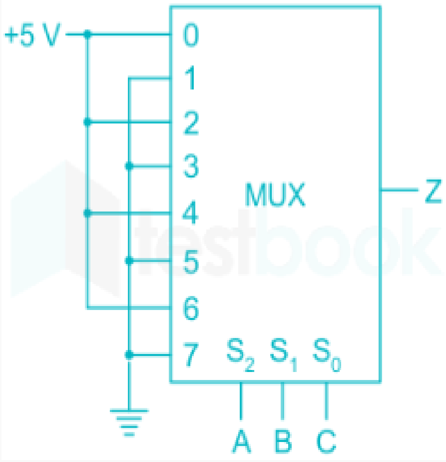
1. Demultiplexer
2. Multiplexer
3. Flip-Flop
4. Counter

57 Which of the following is the correct realization of Z = AB + BC + CA ?

1.


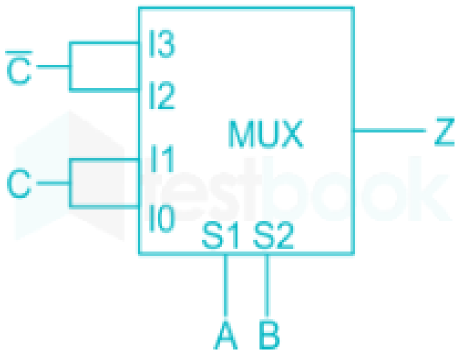
2.


3.


4.


Combinational and Arithmetic Circuits

58 A 4 × 1 Multiplexer is shown in the Figure below. The output Z is



1. A NOR C
2. B NOR C
3. B XOR C
4. A XOR C

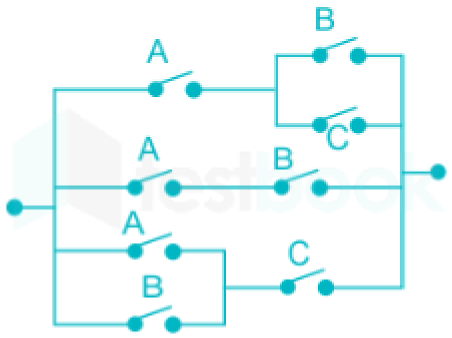
59 Digital multiplexer is basically a combinational logic circuit to perform the operation

1. AND-AND
2. OR-OR
3. AND-OR
4. OR-AND

Combinational and Arithmetic Circuits

60

The minimum Boolean expression for the following circuit is



1. AB + AC + BC
2. A + BC
3. A + B
4. A + B + C

61

Consider the following combinational function block involving four Boolean variables x, y, a, b where x, a, b are inputs and y is the output.

```
f (x, y, a, b)
{
        if (x is 1) y = a;
        else y = b;
}
```

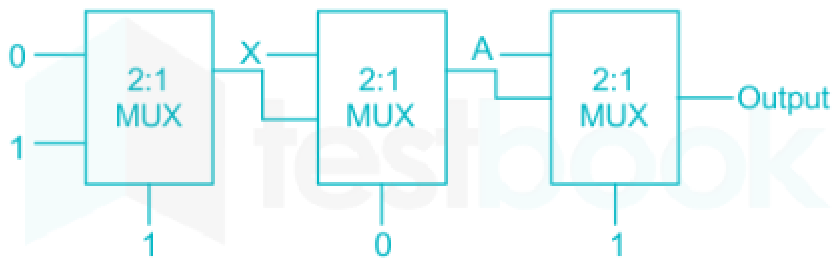
Which one of the following digital logic blocks is the most suitable for implementing this function?

1. Full adder
2. Priority encoder
3. Multiplexer
4. Flip-flop

Combinational and Arithmetic Circuits

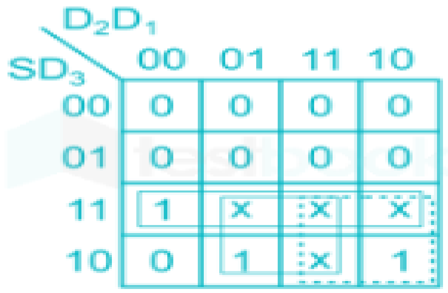
62

The output of the circuit shown is Fig is _______



1. 0
2. X
3. A
4. 1

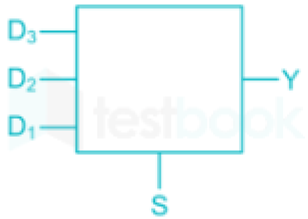
S	D3	D2	D1	Y
0	x	x	x	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	1
1	0	1	1	x
1	1	0	1	x



$Y = SD_3 + SD_1 + SD_2$

63

The combinational circuit shown in the below figure has the function. When selector input 's' is high the circuit is to detect if one of the data lines has logic '1' and no more than one data line is having logic '1', when selector input 'S' is low, the circuit will output '0', regardless of what is on the data lines. The logic of output Y is



1. $S(D_1 + D_2 + D_3)$
2. $S\left(\bar{D}_1 + D_2 + D_3\right)$
3. $\bar{S}(D_1 + D_2) + SD_3$
4. $(D_1 + D_3) + \bar{S}D_2$

THANK YOU